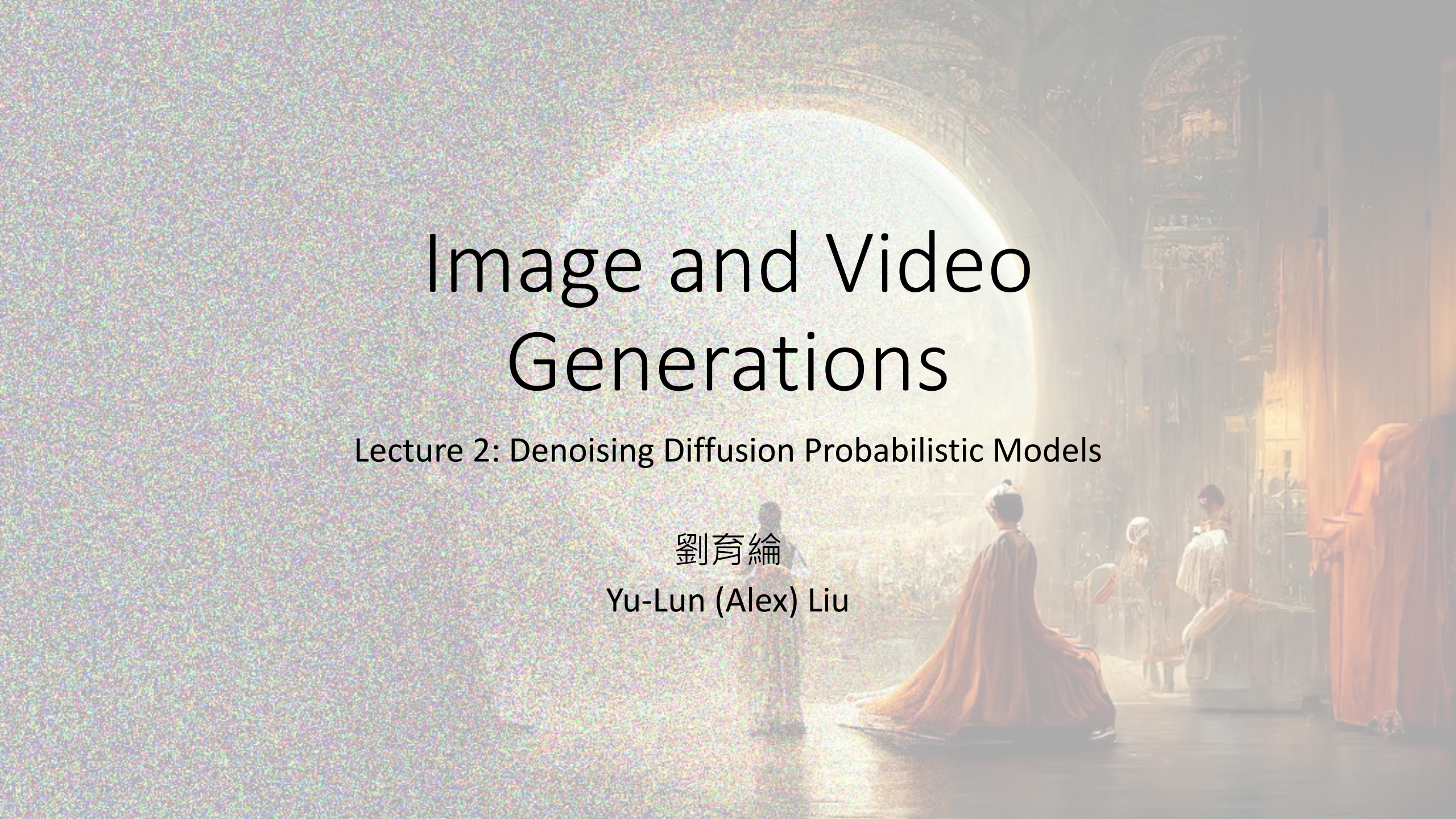


Image and Video Generations

Lecture 2: Denoising Diffusion Probabilistic Models

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Week	Date	Topic	Assignments
1	2025-09-04		
2	2025-09-11	GAN / VAE	
3	2025-09-18	DDPM	#1 – DDPM
4	2025-09-25	DDIM	
5	2025-10-02	CFG / Latent Diffusion / ControlNet / LoRA / Zero-Shot Applications (attending ICCV 2025)	#2 – DDIM & LoRA
6	2025-10-09	DDIM Inversion / Score Distillation	
7	2025-10-16	Diffusion Synchronization / Inverse Problems	#3 – Distillation
8	2025-10-23	Probability Flow ODE / DPM-Solver (attending ICCV 2025)	
9	2025-10-30	Flow Matching	#4 – Flow Matching
10	2025-11-06	Final Project Proposal	
11	2025-11-13		
12	2025-11-20		
13	2025-11-27	Paper Presentation	
14	2025-12-04	Guest Lecture – Ta-Ying (Tim) Cheng	
15	2025-12-11	Guest Lecture – Chieh (Hubert) Lin	
16	2025-12-18	Guest Lecture – Chih-Hao Lin	
17	2025-12-25	Final Project Presentation	

Denoising Diffusion Probabilistic Models (DDPM)

Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020.

Denoising Diffusion Probabilistic Models

Consider a special case of the Markovian hierarchical VAEs where:

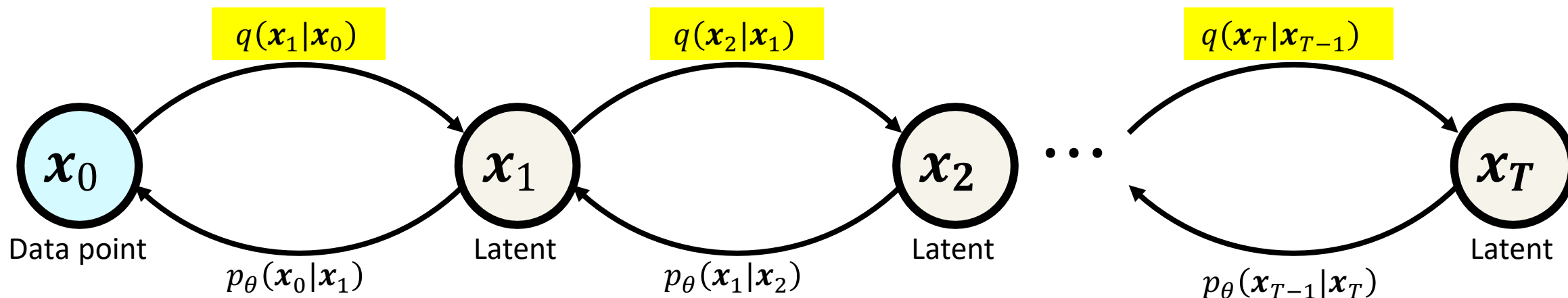
- the latent **dimension** is the **same** as the data dimension, and
- the **variational posteriors** $q_{\phi}(\mathbf{x}_{t+1}|\mathbf{x}_t)$ are not learned but **predefined**:

$$q_{\phi}(\mathbf{x}_{t+1}|\mathbf{x}_t) \rightarrow q(\mathbf{x}_{t+1}|\mathbf{x}_t)$$

Terminology

Forward process (predefined):

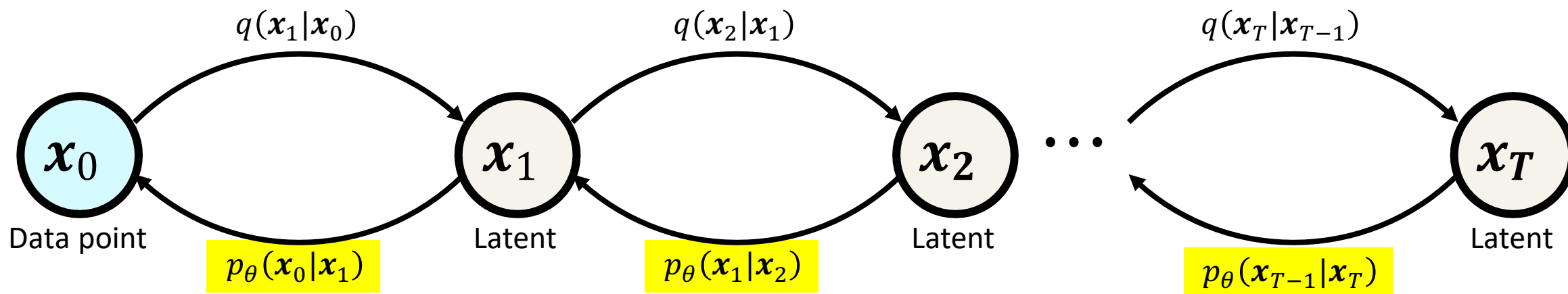
$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$



Terminology

Reverse process (**learned**):

$$p_{\theta}(\mathbf{x}_0|\mathbf{x}_{1:T}) = p_{\theta}(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$



Forward Process (Data $\rightarrow$ Latent) 正向加噪

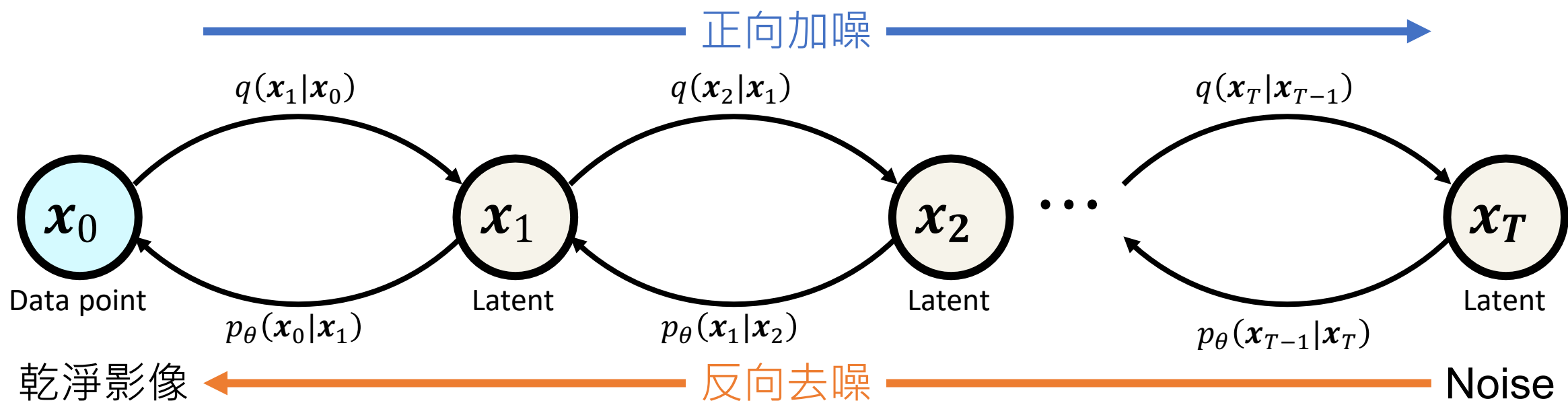
In the forward process, the transition distribution $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ is specifically predefined as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

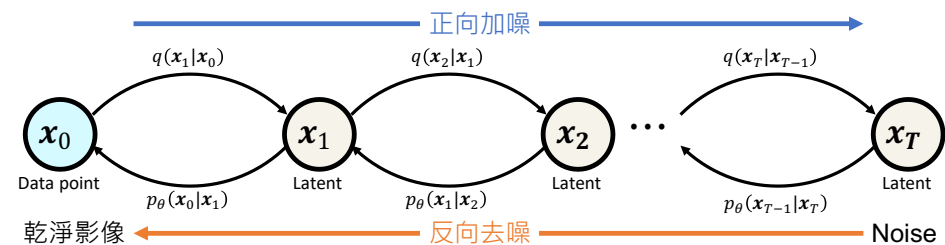
Time-dependent
parameter

where $\{\beta_t \in (0,1)\}_{t=1}^T$ and $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$.

“Adding Gaussian noise iteratively!”



VP-SDE vs. VE-SDE



- $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$,
is called **Variance Preserving (VP)** form.

- There are **other options**. For example:

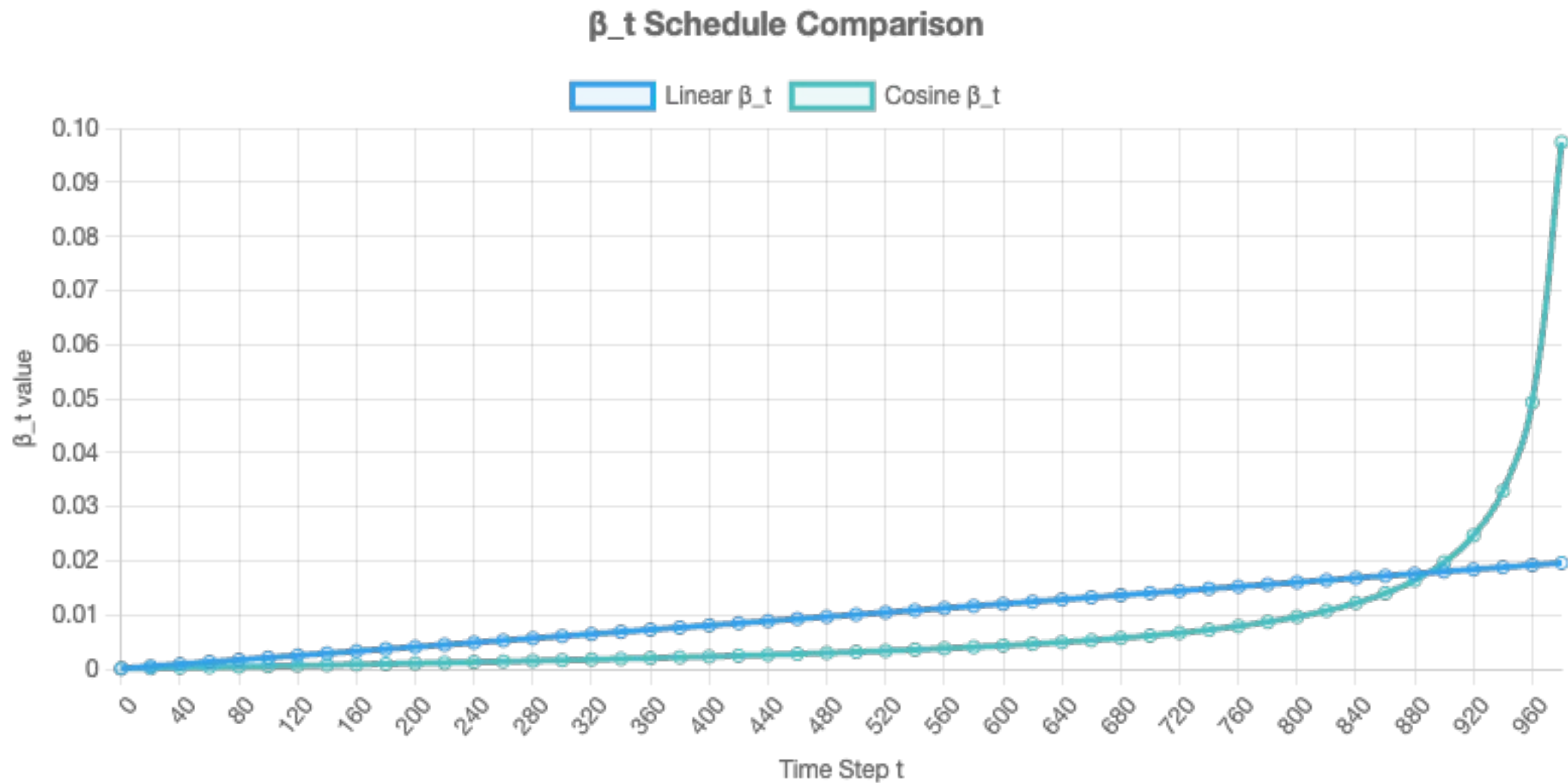
$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \mathbf{x}_{t-1}, (\sigma_i^2 - \sigma_{i-1}^2)\mathbf{I}),$$

which is called **Variance Exploding** form.

Choice of β_t

- Learned.
- Constant.
- Linearly or quadratically increased.
- Follows a **cosine** function
(Nichol and Dhariwal, Improved Denoising Diffusion Probabilistic Models, ICML 2021).

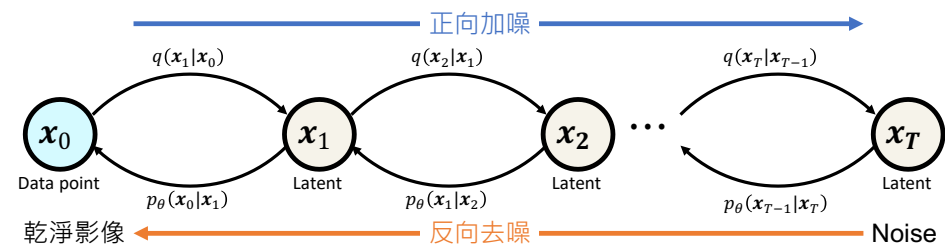
Choice of β_t



How to maximize ELBO in this case?

Disclaimer: We'll skip some complicated equations in the following slides.

ELBO



How to minimize the *negative* ELBO in this case?

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$= \dots =$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

$$+\mathbb{E}_{q(\mathbf{x}_{T-1}|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_T|\mathbf{x}_{T-1}) \| p(\mathbf{x}_T))]$$

$$+\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1}|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_t|\mathbf{x}_{t-1}) \| p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1}))]$$

ELBO: Complex Derivations

$$\log p(\mathbf{x}) = \log \int p(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \quad (34)$$

$$= \log \int \frac{p(\mathbf{x}_{0:T}) q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} d\mathbf{x}_{1:T} \quad (35)$$

$$= \log \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right] \quad (36)$$

$$\geq \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right] \quad (37)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)}{\prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \quad (38)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1) \prod_{t=2}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)}{q(\mathbf{x}_T | \mathbf{x}_{T-1}) \prod_{t=1}^{T-1} q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \quad (39)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1) \prod_{t=1}^{T-1} p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1})}{q(\mathbf{x}_T | \mathbf{x}_{T-1}) \prod_{t=1}^{T-1} q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \quad (40)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)}{q(\mathbf{x}_T | \mathbf{x}_{T-1})} \right] + \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \prod_{t=1}^{T-1} \frac{p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1})}{q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \quad (41)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)] + \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T | \mathbf{x}_{T-1})} \right] + \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\sum_{t=1}^{T-1} \log \frac{p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1})}{q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \quad (42)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)] + \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T | \mathbf{x}_{T-1})} \right] + \sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1})}{q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \quad (43)$$

$$= \mathbb{E}_{q(\mathbf{x}_1 | \mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)] + \mathbb{E}_{q(\mathbf{x}_{T-1}, \mathbf{x}_T | \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T | \mathbf{x}_{T-1})} \right] + \sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1} | \mathbf{x}_0)} \left[\log \frac{p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1})}{q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \quad (44)$$

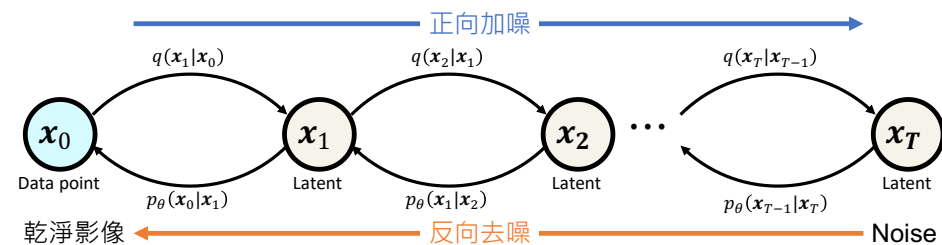
$$= \underbrace{\mathbb{E}_{q(\mathbf{x}_1 | \mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{\mathbb{E}_{q(\mathbf{x}_{T-1} | \mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_{T-1}) \parallel p(\mathbf{x}_T))]}_{\text{prior matching term}} \quad (45)$$

$$- \sum_{t=1}^{T-1} \underbrace{\mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1} | \mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_t | \mathbf{x}_{t-1}) \parallel p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1}))]}_{\text{consistency term}}$$

<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

<https://zhuanlan.zhihu.com/p/558937247>

ELBO



How to minimize the *negative* ELBO in this case?

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$= \dots =$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

Reconstruction
term

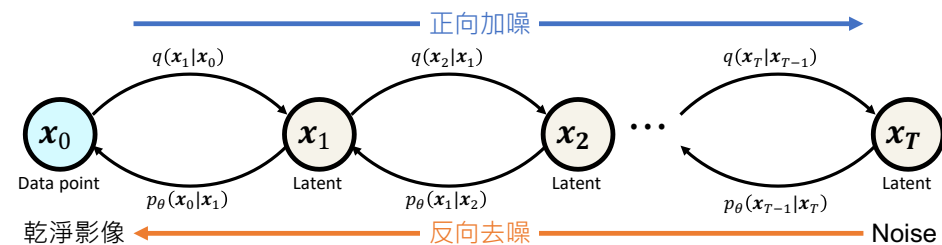
$$+\mathbb{E}_{q(\mathbf{x}_{T-1}|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_T|\mathbf{x}_{T-1})\|p(\mathbf{x}_T))]$$

Prior matching
term

$$+\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1}|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_t|\mathbf{x}_{t-1})\|p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1}))]$$

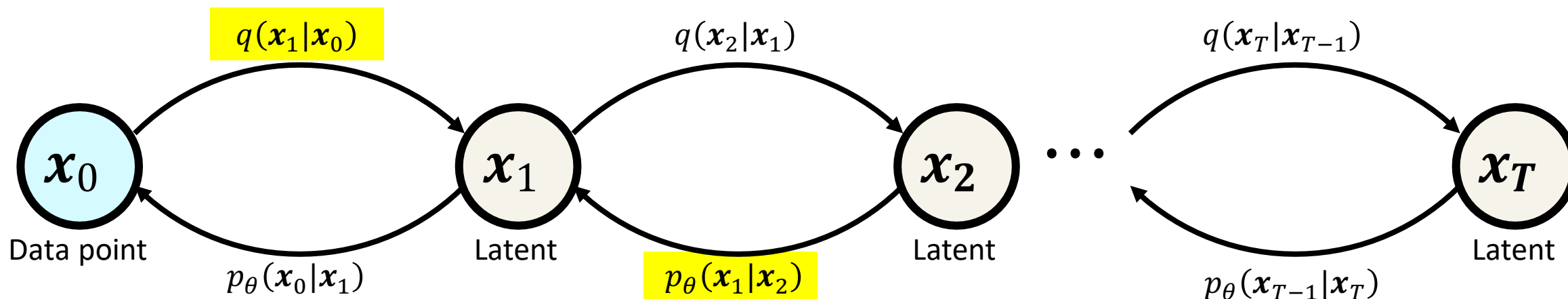
Consistency
term

Consistency Term



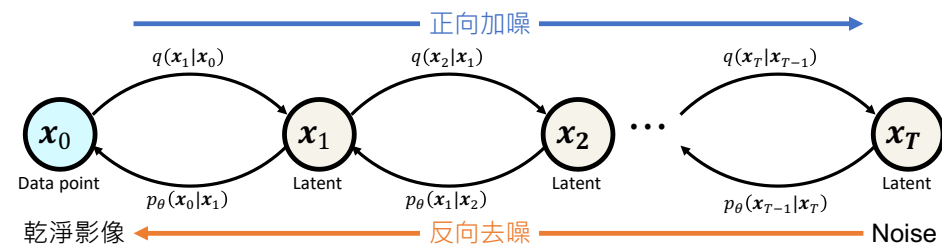
$$\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1}|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_t|\mathbf{x}_{t-1}) || p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1}))]$$

Make the forward and reverse steps be consistent at each time step.



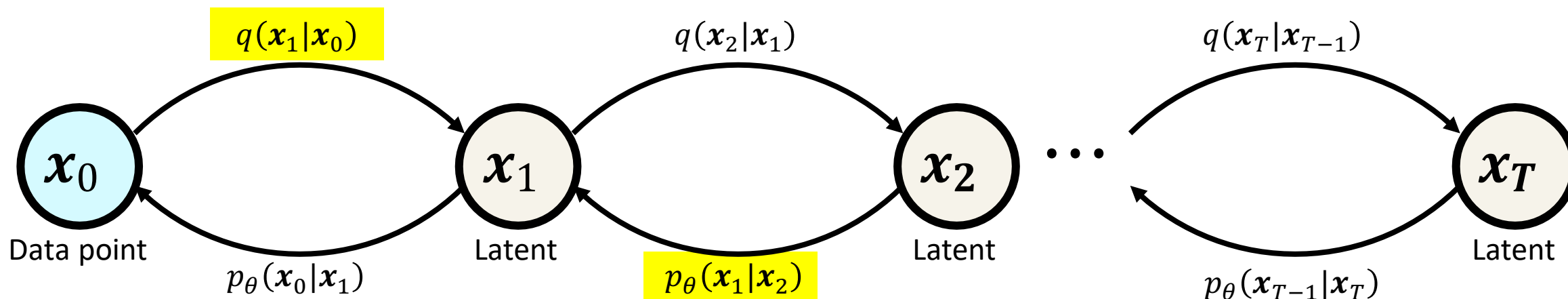
這兩個到達 x_t 的路徑要一致

Consistency Term



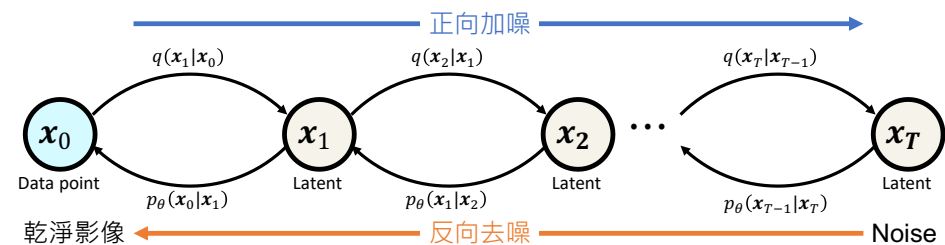
$$\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1} | \mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_t | \mathbf{x}_{t-1}) || p_\theta(\mathbf{x}_t | \mathbf{x}_{t+1}))]$$

Expectation over two random variables; computationally expensive.



這兩個到達 x_t 的路徑要一致

ELBO



Can we avoid having two random variables in an expectation?

Let's re-decompose the ELBO using the fact that

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0).$$

Q. Why $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)$?

馬可夫過程的條件機率僅僅與系統的當前狀態相關，而與它的過去歷史或未來狀態，都是獨立、不相關的

ELBO: Complex Derivations

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) = \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) q(\mathbf{x}_t | \mathbf{x}_0)}{q(\mathbf{x}_{t-1} | \mathbf{x}_0)} \quad (46)$$

ELBO: Complex Derivations

$$\log p(\mathbf{x}) \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \quad (47)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] \quad (48)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) \prod_{t=2}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0) \prod_{t=2}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] \quad (49)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) \prod_{t=2}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0) \prod_{t=2}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)} \right] \quad (50)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p_{\theta}(\mathbf{x}_T) p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)} \right] \quad (51)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}} \right] \quad (52)$$

ELBO: Complex Derivations

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}} \right] \quad (53)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)}{\cancel{q(\mathbf{x}_1|\mathbf{x}_0)}} + \log \frac{\cancel{q(\mathbf{x}_1|\mathbf{x}_0)}}{q(\mathbf{x}_T|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \quad (54)$$

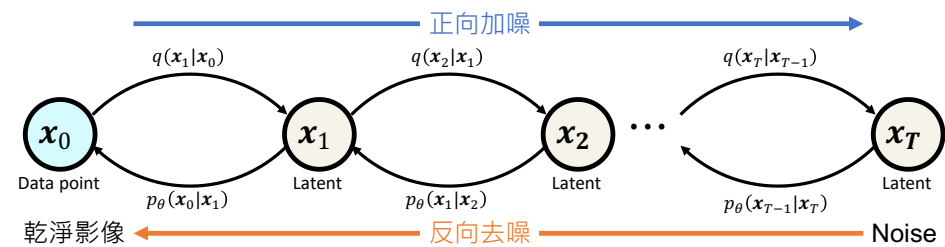
$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^T \log \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \quad (55)$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)] + \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} \right] + \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \quad (56)$$

$$= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)] + \mathbb{E}_{q(\mathbf{x}_T|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} \right] + \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_{t-1}|\mathbf{x}_0)} \left[\log \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \quad (57)$$

$$= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}} \quad (58)$$

ELBO



Decompose the negative ELBO in a **different** way:

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$= \dots =$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

Reconstruction
term $\mathcal{L}_0$

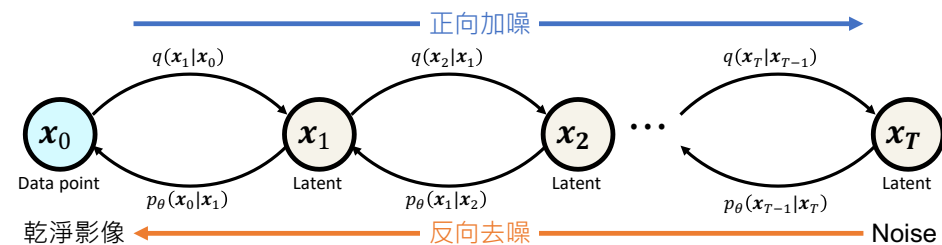
$$+D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T))$$

New prior matching
term $\mathcal{L}_T$

$$+\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Denoising
matching
term $\mathcal{L}_{t-1}$

Reconstruction Term $\mathcal{L}_0$

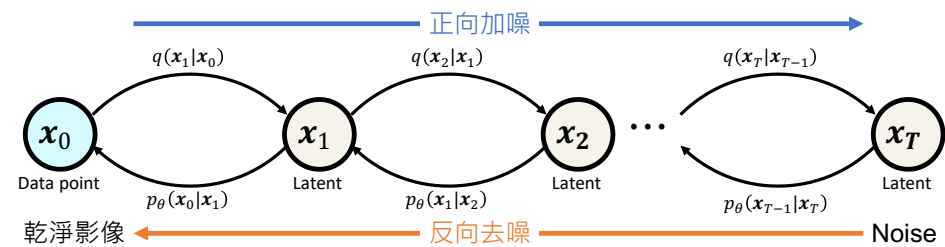


$$-\mathbb{E}_{q(x_1|x_0)}[\log p_\theta(x_0|x_1)]$$

下標表示的是對哪個隨機變數取期望值，以及這個隨機變數的分佈
期望值下標 $q(x_1|x_0)$ 代表這個 loss 是對所有可能的 noise x_1 取平均
而 x_1 的分佈由 forward process $q(x_1|x_0)$ 決定

The same loss term in VAE, but applied only to the **final** reverse step.

ELBO



Decompose the negative ELBO in a **different** way:

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$= \dots =$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

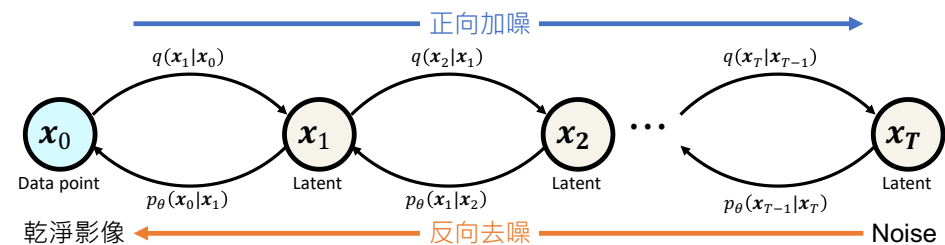
$$+ D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T))$$

New prior matching
term $\mathcal{L}_T$

$$+ \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Prior Matching Term $\mathcal{L}_T$

Prior Matching Term $\mathcal{L}_T$



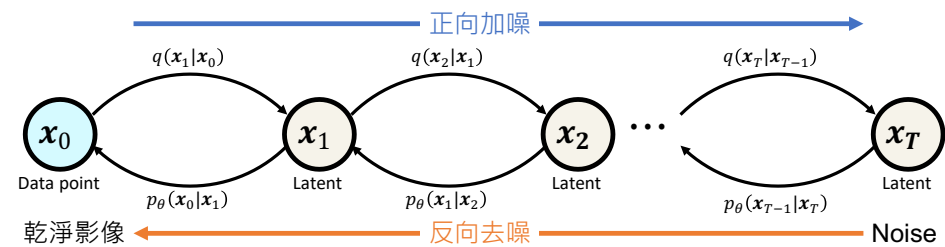
$$D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))$$

- Identical to the KL divergence term in VAE.
- Note that there is nothing to be optimized; $q(\mathbf{x}_T|\mathbf{x}_0)$ and $p(\mathbf{x}_T)$ are predefined.

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

$$p(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$$

Forward Convergence



Then, $q(\mathbf{x}_T|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$?

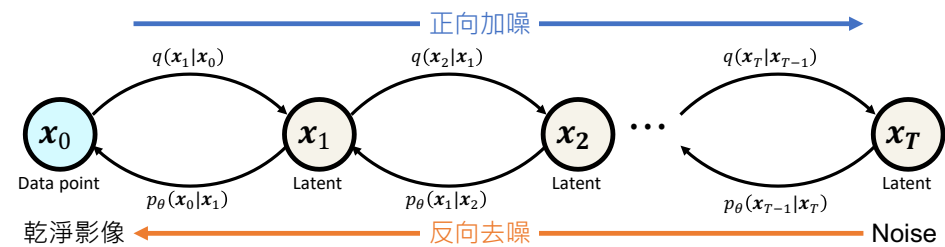
Yes, under certain assumptions.

Recall

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

where $\{\beta_t \in (0,1)\}_{t=1}^T$ and $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$.

Forward Convergence



Then, $q(\mathbf{x}_T|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$?

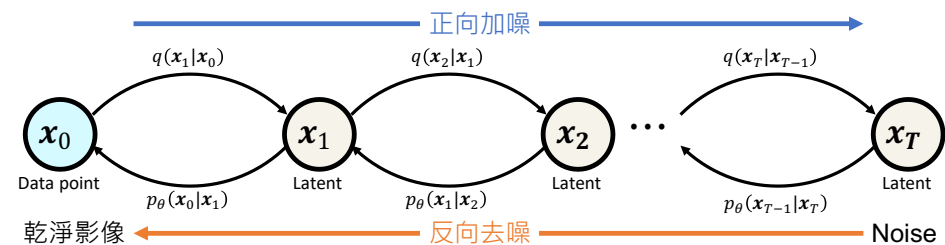
Yes, under certain assumptions.

Let $\alpha_t = 1 - \beta_t$.

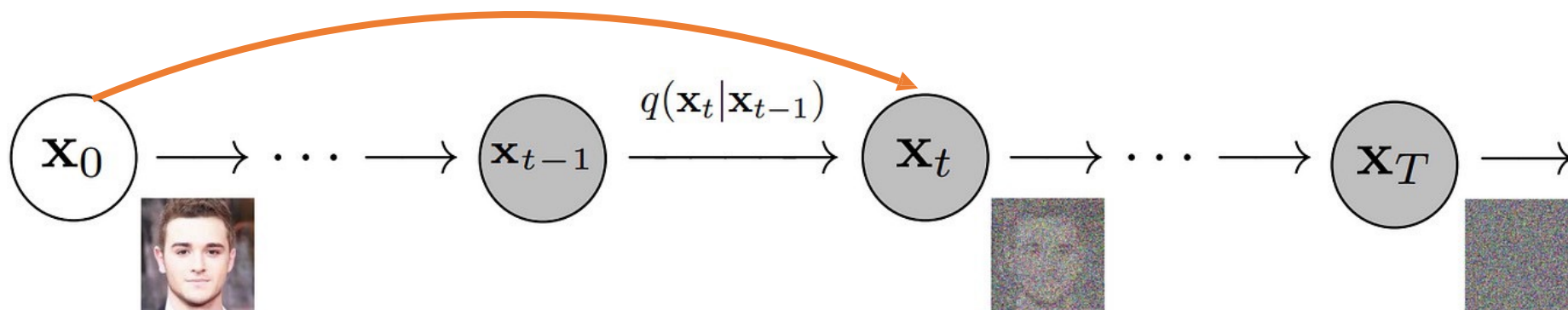
$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

where $\{\alpha_t \in (0,1)\}_{t=1}^T$ and $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_T$.

$$q(\mathbf{x}_t | \mathbf{x}_0)$$



Can we derive $q(\mathbf{x}_t | \mathbf{x}_0)$ from the sequence of $q(\mathbf{x}_{t'} | \mathbf{x}_{t'-1})$ for $t = 1, \dots, t'$?



$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

Basics: Combination of Gaussian Variables

Suppose $\mathbf{x}_1 \sim \mathcal{N}(\mu_1, \sigma_1^2 \mathbf{I})$ and $\mathbf{x}_2 \sim \mathcal{N}(\mu_2, \sigma_2^2 \mathbf{I})$

Q. What is the distribution of $\mathbf{x}_1 + \mathbf{x}_2$?

Basics: Combination of Gaussian Variables

Suppose $\mathbf{x}_1 \sim \mathcal{N}(\mu_1, \sigma_1^2 \mathbf{I})$ and $\mathbf{x}_2 \sim \mathcal{N}(\mu_2, \sigma_2^2 \mathbf{I})$

A. $\mathbf{x}_1 + \mathbf{x}_2 \sim \mathcal{N}(\mu_1 + \mu_2, (\sigma_1^2 + \sigma_2^2) \mathbf{I})$

Basics: Combination of Gaussian Variables

Suppose $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and

$\boldsymbol{x}_1 = \sigma_1 \boldsymbol{\varepsilon}_1$ and $\boldsymbol{x}_2 = \sigma_2 \boldsymbol{\varepsilon}_2$.

Q. What is the distribution of $\boldsymbol{x}_1 + \boldsymbol{x}_2$?

Basics: Combination of Gaussian Variables

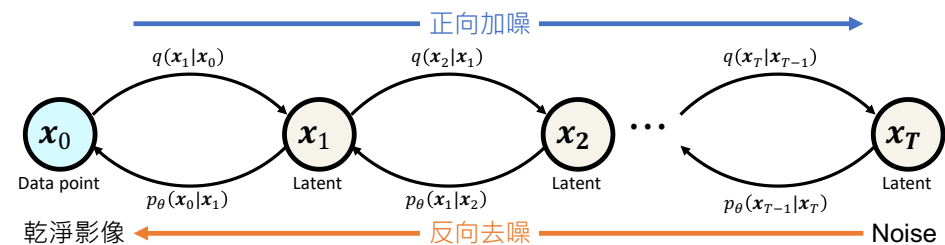
Suppose $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and

$\boldsymbol{x}_1 = \sigma_1 \boldsymbol{\varepsilon}_1$ and $\boldsymbol{x}_2 = \sigma_2 \boldsymbol{\varepsilon}_2$.

A. $\boldsymbol{x}_1 + \boldsymbol{x}_2 \sim \mathcal{N}(\mathbf{0}, (\sigma_1^2 + \sigma_2^2)\mathbf{I})$.

$\boldsymbol{x}_1 + \boldsymbol{x}_2 = \sqrt{\sigma_1^2 + \sigma_2^2} \boldsymbol{\varepsilon}$ where $\boldsymbol{\varepsilon}$ is another standard normal sample.

Forward Convergence



$$q(x_1|x_0) = \mathcal{N}(x_1; \sqrt{\alpha_1}x_0, (1 - \alpha_1) \mathbf{I})$$

$$q(x_2|x_1) = \mathcal{N}(x_2; \sqrt{\alpha_2}x_1, (1 - \alpha_2) \mathbf{I})$$

Q. What is the distribution of $q(x_2|x_0)$?

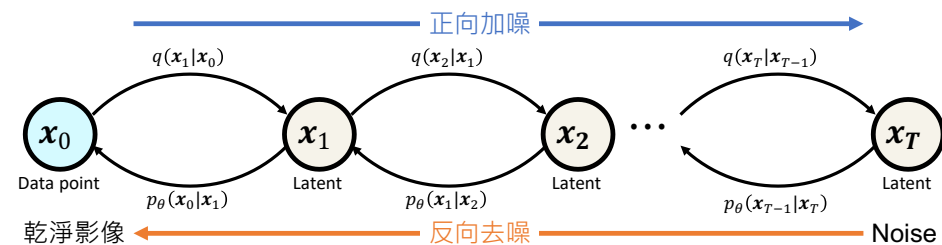
Hint. Let's use the reparameterization trick:

$$x_1 = \sqrt{\alpha_1}x_0 + \sqrt{1 - \alpha_1}\epsilon_0$$

$$x_2 = \sqrt{\alpha_2}x_1 + \sqrt{1 - \alpha_2}\epsilon_1$$

$$\epsilon_0, \epsilon_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Forward Convergence



$$\mathbf{A.} \quad \mathbf{x}_2 = \sqrt{\alpha_2} \mathbf{x}_1 + \sqrt{1 - \alpha_2} \boldsymbol{\epsilon}_1$$

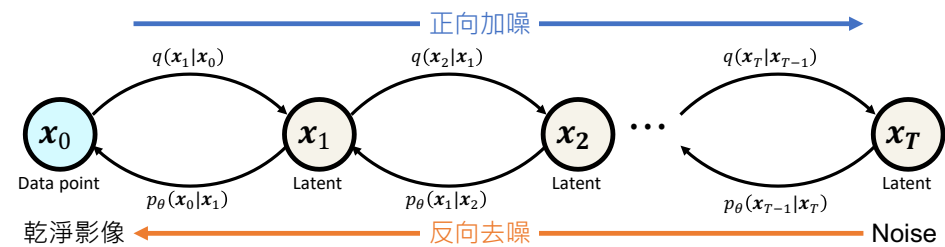
$$= \sqrt{\alpha_2} (\sqrt{\alpha_1} \mathbf{x}_0 + \sqrt{1 - \alpha_1} \boldsymbol{\epsilon}_0) + \sqrt{1 - \alpha_2} \boldsymbol{\epsilon}_1$$

$$= \sqrt{\alpha_2 \alpha_1} \mathbf{x}_0 + \sqrt{\alpha_2 (1 - \alpha_1)} \boldsymbol{\epsilon}_0 + \sqrt{1 - \alpha_2} \boldsymbol{\epsilon}_1$$

$$= \sqrt{\alpha_2 \alpha_1} \mathbf{x}_0 + \sqrt{(1 - \alpha_2 \alpha_1)} \bar{\boldsymbol{\epsilon}}_0 \quad \mathbf{x}_1 + \mathbf{x}_2 \sim \mathcal{N}(\mathbf{0}, (\sigma_1^2 + \sigma_2^2) \mathbf{I})$$

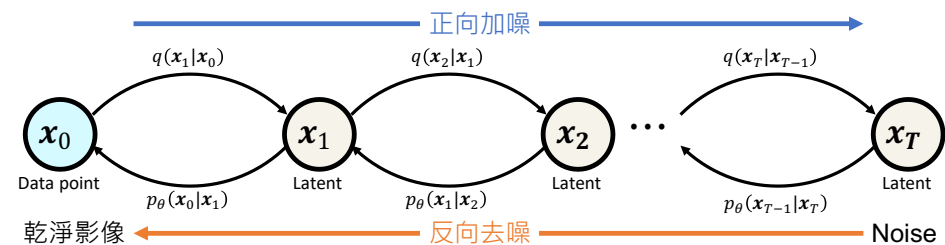
$$\therefore q(\mathbf{x}_2 | \mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_2 \alpha_1} \mathbf{x}_0, (1 - \alpha_2 \alpha_1) \mathbf{I})$$

Forward Convergence

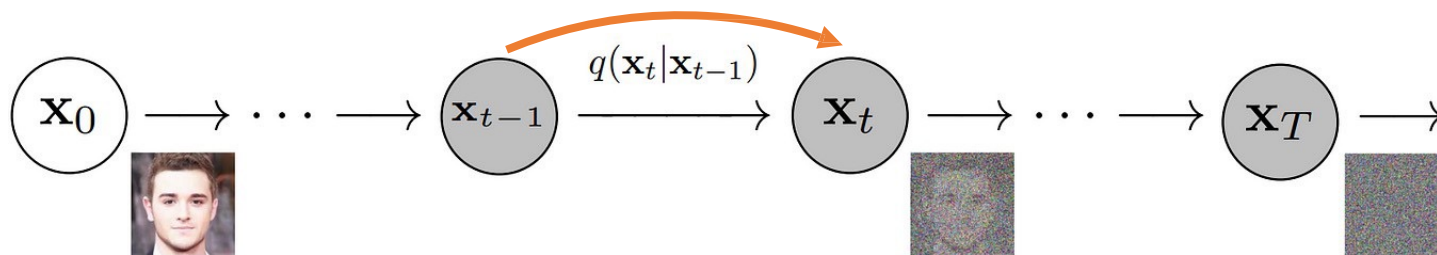


$$\begin{aligned}
 \mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\
 &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \boldsymbol{\epsilon}_{t-2}) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\
 &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \boldsymbol{\epsilon}_{t-2} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\
 &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{(1 - \alpha_t \alpha_{t-1})} \bar{\boldsymbol{\epsilon}}_{t-2} \\
 &= \dots \\
 &= \sqrt{\sum_{i=1}^t \alpha_i} \mathbf{x}_0 + \sqrt{(1 - \prod_{i=1}^t \alpha_i)} \bar{\boldsymbol{\epsilon}}_0
 \end{aligned}$$

Forward Convergence

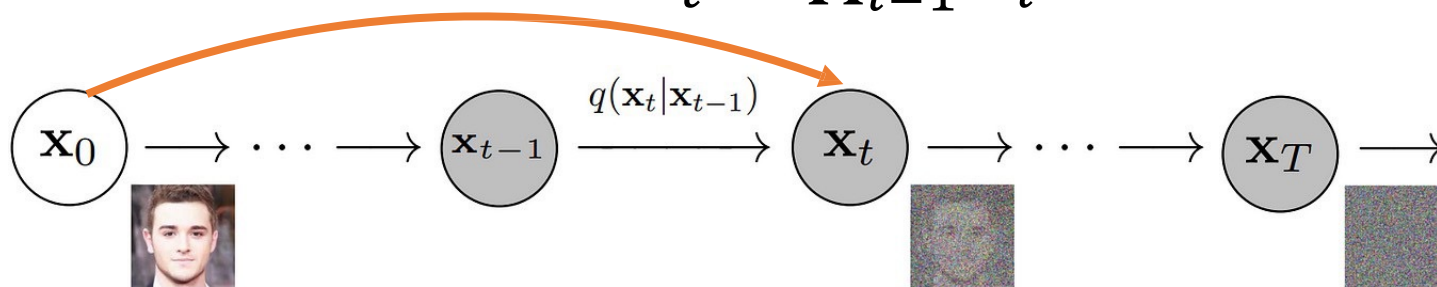


$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)\mathbf{I})$$



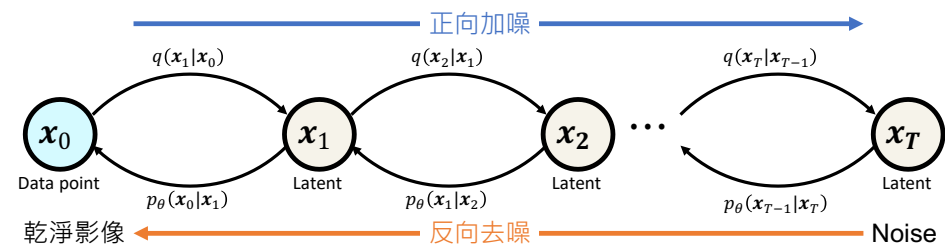
$$q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ *Also a normal distribution!*



Forward jump/正向跳躍

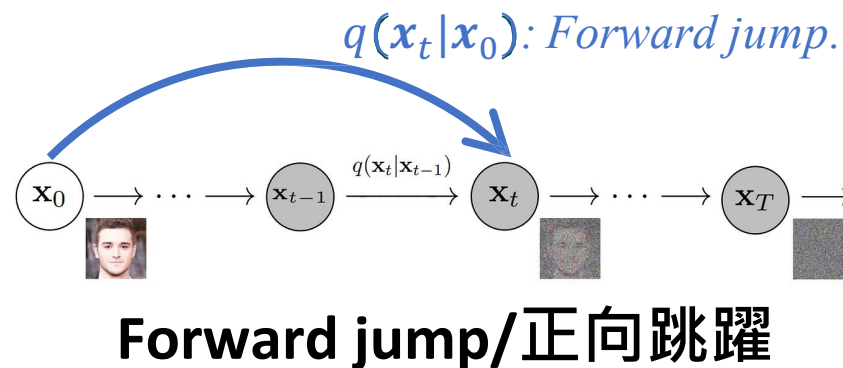
$$q(\mathbf{x}_t | \mathbf{x}_0)$$



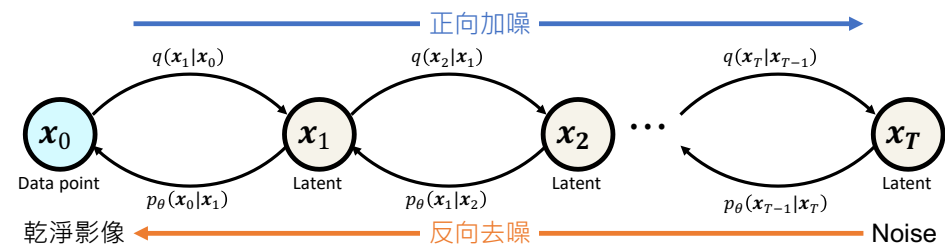
Given $\mathbf{x}_0$, $\mathbf{x}_t$ at any arbitrary timestep t can be directly sampled from a Gaussian distribution without a Markov chain:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}).$$

Note that $\bar{\alpha}_1 > \bar{\alpha}_2 > \dots > \bar{\alpha}_T$.



Forward Convergence



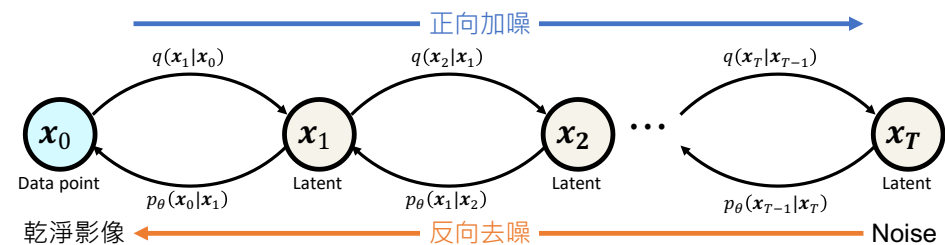
$$q(\mathbf{x}_T | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_T; \sqrt{\bar{\alpha}_T} \mathbf{x}_0, (1 - \bar{\alpha}_T) \mathbf{I})$$

$$\text{where } \bar{\alpha}_T = \prod_{t=1}^T \alpha_t = \prod_{t=1}^T (1 - \beta_t).$$

Q. When $\{\beta_t \in (0,1)\}_{t=1}^T$, What is

$$\begin{aligned} \lim_{T \rightarrow \infty} \bar{\alpha}_T &= \lim_{T \rightarrow \infty} \prod_{t=1}^T (1 - \beta_t)? \\ &= 0 \end{aligned}$$

Forward Convergence

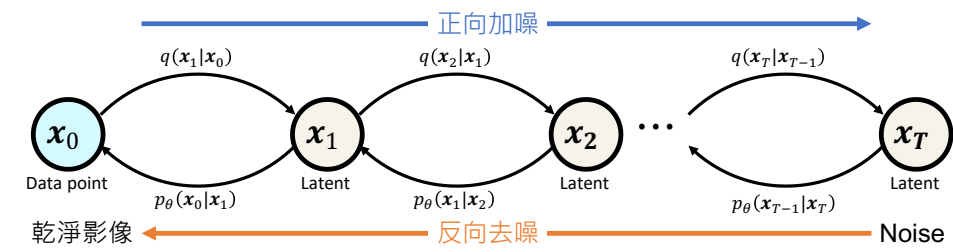


$$q(\mathbf{x}_T | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_T; \sqrt{\bar{\alpha}_T} \mathbf{x}_0, (1 - \bar{\alpha}_T) \mathbf{I})$$

$$\text{where } \bar{\alpha}_T = \prod_{t=1}^T \alpha_t = \prod_{t=1}^T (1 - \beta_t).$$

As $T \rightarrow \infty$, $q(\mathbf{x}_T | \mathbf{x}_0)$ converges to the standard normal distribution $\mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$.

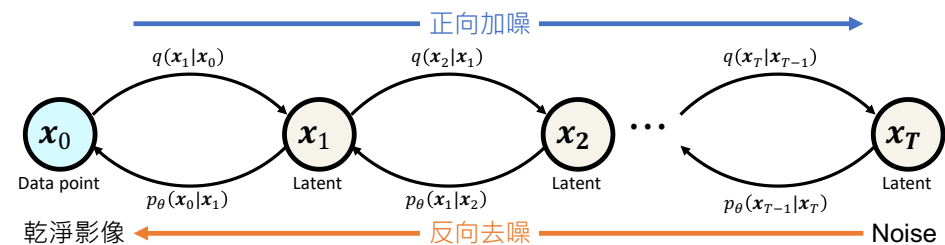
Prior Matching Term $\mathcal{L}_T$



$$D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))$$

Close to zero by the definition of the forward transition distribution $q(\mathbf{x}_t|\mathbf{x}_{t-1})$. *Nothing to do for the optimization.*

ELBO



Decompose the negative ELBO in a **different** way:

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$= \dots =$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

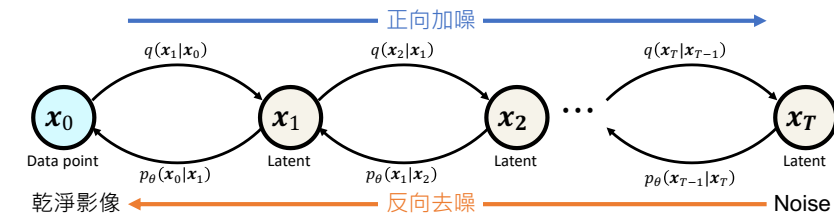
$$+ \cancel{D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T))} \quad 0$$

$$+ \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

**Denoising
matching
term $\mathcal{L}_{t-1}$**

Denoising Matching Term $\mathcal{L}_{t-1}$

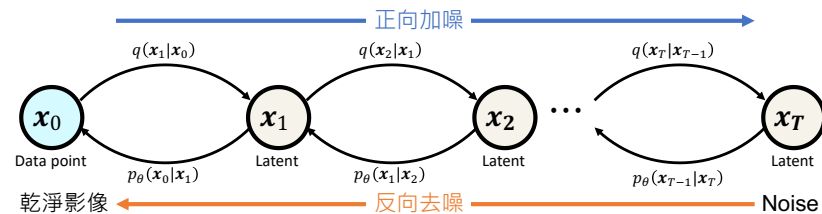
Denoising Matching Term $\mathcal{L}_{t-1}$



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

The **variational** distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ should be close to $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ for each t .

Denoising Matching Term $\mathcal{L}_{t-1}$

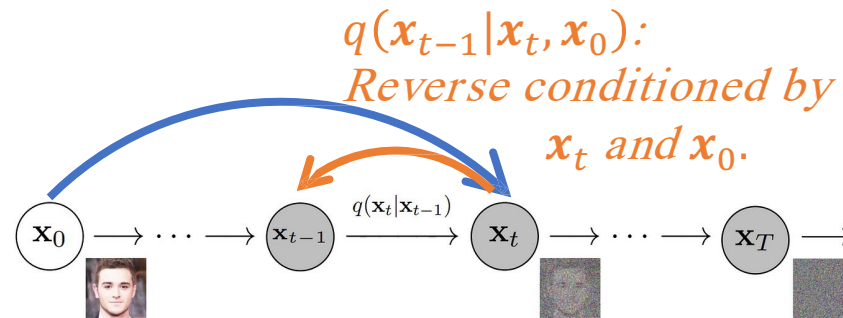


What is $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$?

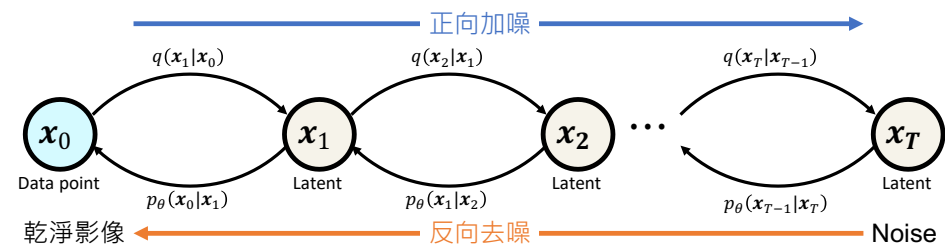
$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

Same as $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, the forward transition, since it's a Markovian process.

We have seen how to compute these.
Forward jump/正向跳躍



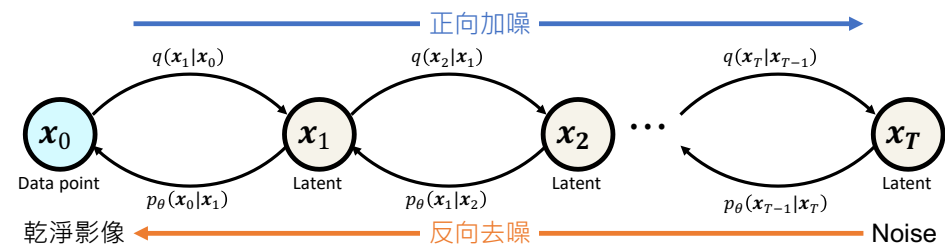
$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$



$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = q(\mathbf{x}_t | \mathbf{x}_{t-1}) \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)}$$

Q. What are $q(\mathbf{x}_t | \mathbf{x}_{t-1})$, $q(\mathbf{x}_{t-1} | \mathbf{x}_0)$, and $q(\mathbf{x}_t | \mathbf{x}_0)$?

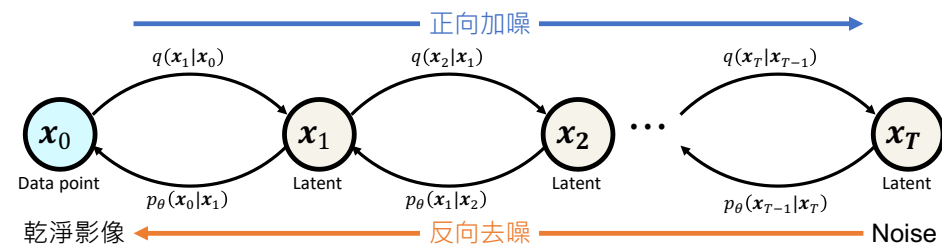
$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$



$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = q(\mathbf{x}_t | \mathbf{x}_{t-1}) \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)}$$

- $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$
- $q(\mathbf{x}_{t-1} | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0, (1 - \bar{\alpha}_{t-1}) \mathbf{I})$
- $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$

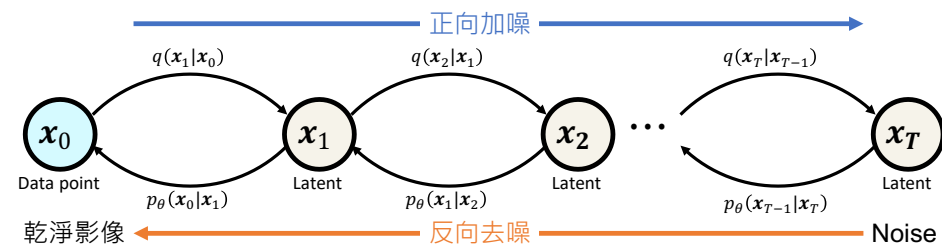
$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$



$$\begin{aligned}
 q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t | \mathbf{x}_{t-1}) \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \\
 &\propto \exp \left(-\frac{1}{2} \left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_{t-1})}{1 - \alpha_t} + \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0)}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)}{1 - \bar{\alpha}_t} \right) \right) \\
 &= \dots \\
 &= \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I}) \quad \text{Another normal distribution!}
 \end{aligned}$$

$$\text{where } \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 \text{ and } \tilde{\sigma}_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

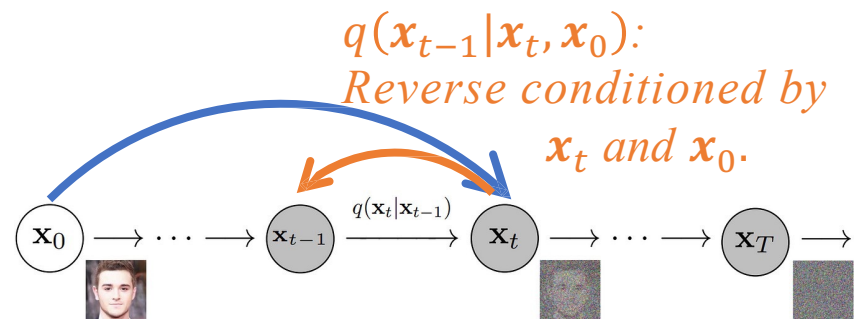
$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$



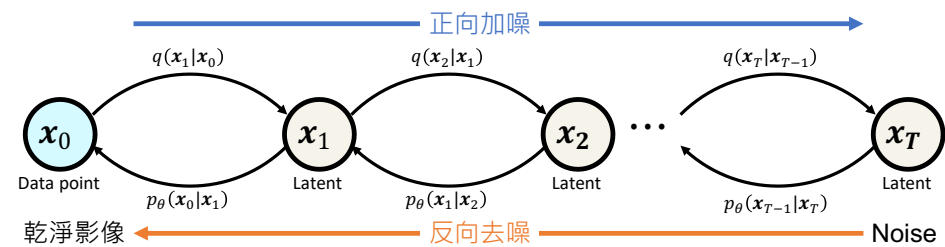
- The **mean** $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0$ is a function of both $\mathbf{x}_t$ and $\mathbf{x}_0$.
 $\mathbf{x}_{t-1}$ 的 mean $\tilde{\mu}$ 是 $\mathbf{x}_t$ 與 $\mathbf{x}_0$ 的線性內插

- The **covariance** $\tilde{\sigma}_t^2 \mathbf{I} = \left(\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t \right) \mathbf{I}$ is **predefined** from the user-defined $\{\beta_t\}_{t=1}^T$.

$\mathbf{x}_{t-1}$ 的 variance $\tilde{\sigma}_t^2$ 與 $\mathbf{x}_t$ 或 $\mathbf{x}_0$ 無關，
 僅由 time step t 與 β_t 預先定義



$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$

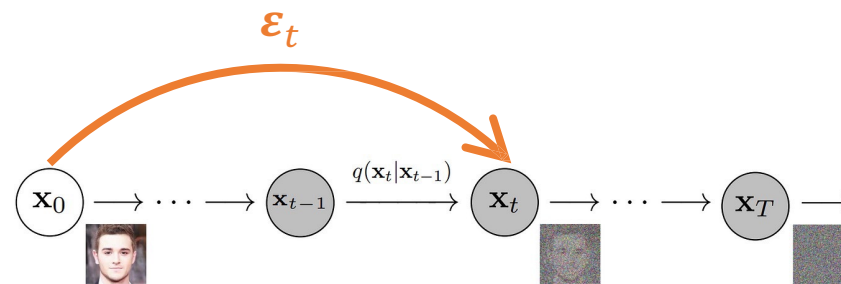


From the forward jump $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$,
 $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}_t$ where $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

If $\mathbf{x}_t$ and $\mathbf{x}_0$ are given, define $\boldsymbol{\varepsilon}_t$ as

$$\boldsymbol{\varepsilon}_t = \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_t}}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{x}_0$$

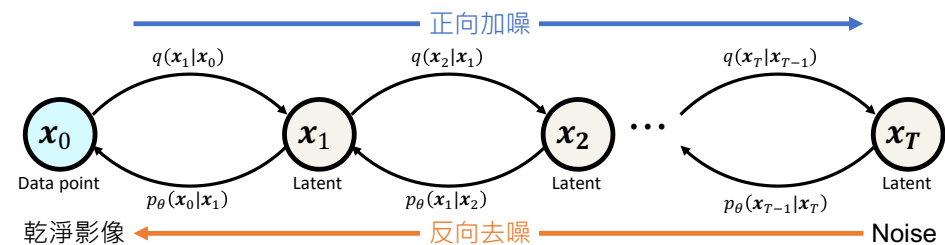
Q. Rewrite $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)$ as a function of $\mathbf{x}_t$ and $\boldsymbol{\varepsilon}_t$.



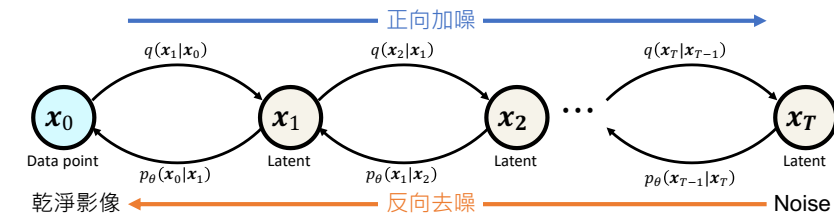
$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$

A.

$$\begin{aligned} \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\varepsilon}_t \right) \end{aligned}$$



Denoising Matching Term $\mathcal{L}_{t-1}$

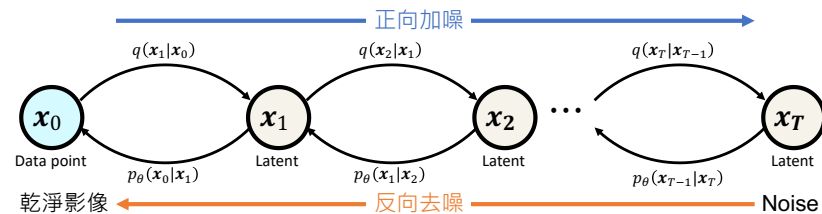


Back to the denoising matching term...

$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

How to model the **variational** distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$?

Denoising Matching Term $\mathcal{L}_{t-1}$

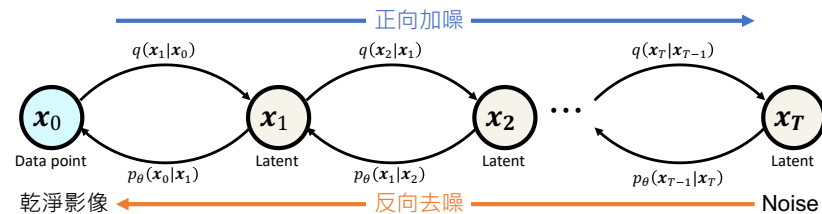


- For $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$, the variance $\tilde{\sigma}_t^2$ is *not* a function of $\mathbf{x}_t$ and $\mathbf{x}_0$.
- Hence, define the variational distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ as

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}),$$

where $\mu_\theta(\mathbf{x}_t, t)$ is the **mean predictor**.

Denoising Matching Term $\mathcal{L}_{t-1}$



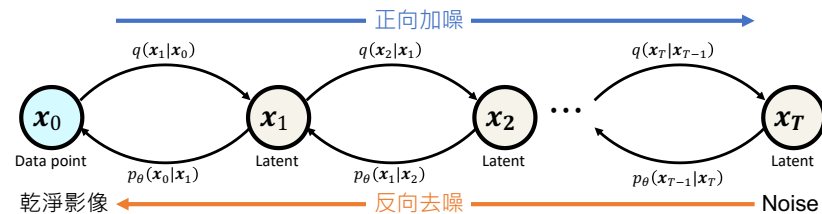
- For $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$, the variance $\tilde{\sigma}_t^2$ is *not* a function of $\mathbf{x}_t$ and $\mathbf{x}_0$.
- Hence, **define** the variational distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ as

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}),$$

where $\mu_\theta(\mathbf{x}_t, t)$ is the mean predictor.

Q. 為什麼我們要把 likelihood (p_θ) 的 variance 定義成與 posterior (q) 相同？

Denoising Matching Term $\mathcal{L}_{t-1}$



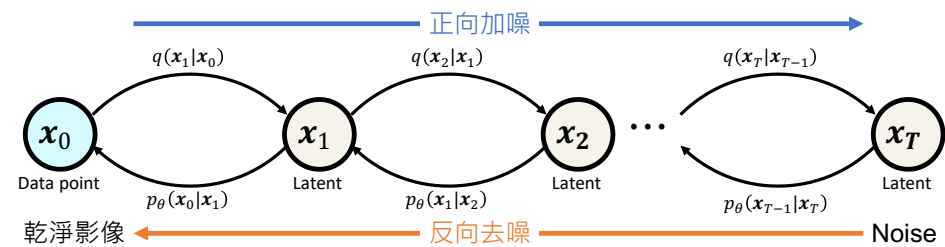
- For $q(x_{t-1}|x_t, x_0) = \mathcal{N}(\tilde{\mu}(x_t, x_0), \tilde{\sigma}_t^2 \mathbf{I})$, the variance $\tilde{\sigma}_t^2$ is *not* a function of x_t and x_0 .
- Hence, **define** the variational distribution $p_\theta(x_{t-1}|x_t)$ as

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t), \tilde{\sigma}_t^2 \mathbf{I}),$$

where $\mu_\theta(x_t, t)$ is the mean predictor.

A. 簡化 optimization 目標：當我們最小化 KL divergence，如果兩個分佈的 variance 相同，KL divergence 會簡化為只需要匹配 mean，這讓訓練變成一個簡單的 mean 預測問題

Denoising Matching Term

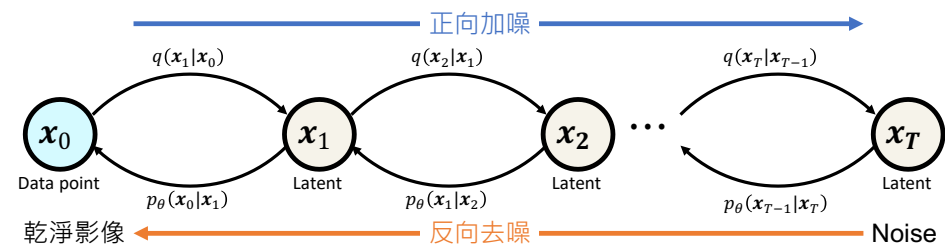


How to compute

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]?$$

Q. When $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_p, \sigma^2 \mathbf{I})$ and $q(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_q, \sigma^2 \mathbf{I})$, What is $D_{KL}(p \| q)$?

Denoising Matching Term



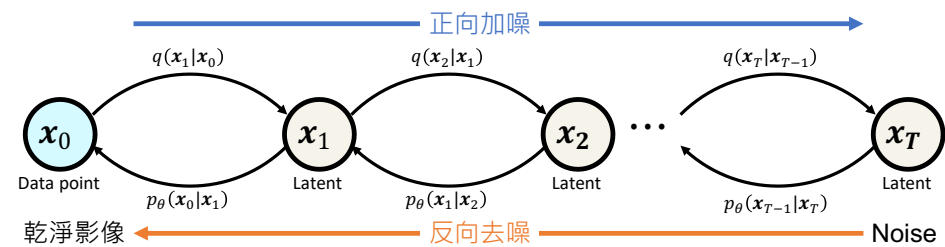
A.

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_p, \sigma^2 \mathbf{I})$$

$$q(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_q, \sigma^2 \mathbf{I})$$

$$D_{KL}(p\|q) = \frac{1}{2\sigma^2} \|\boldsymbol{\mu}_q - \boldsymbol{\mu}_p\|^2$$

Denoising Matching Term



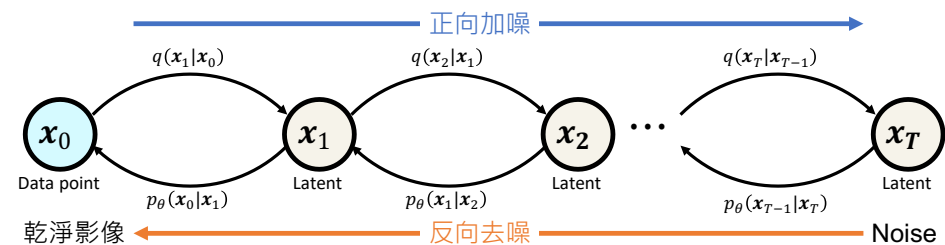
How to compute

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]?$$

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$= \frac{1}{2\tilde{\sigma}_t^2} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[\|\mu_{\theta}(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2]$$

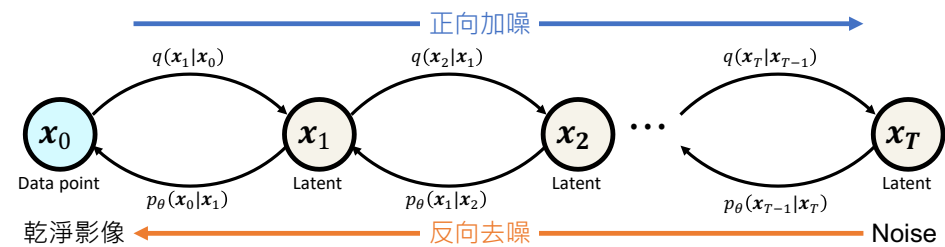
x_0 Predictor



Q. What if we have a x_0 predictor $\hat{x}_\theta(x_t, t)$ instead of the mean predictor $\mu_\theta(x_t, t)$? Note that

$$\tilde{\mu}(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0$$

$\mathbf{x}_0$ Predictor



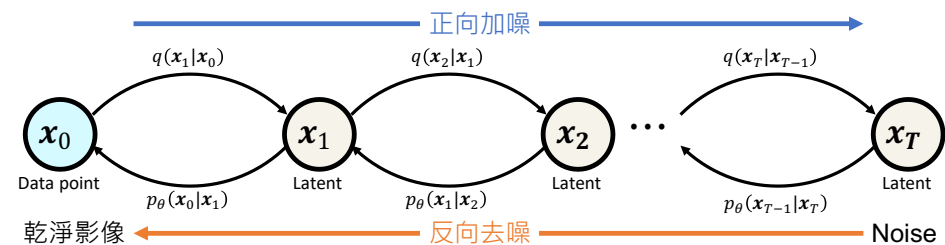
$$\mathbf{A.} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$= \frac{1}{2\tilde{\sigma}_t^2} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[\|\mu_{\theta}(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2]$$

$$= \frac{1}{2\tilde{\sigma}_t^2} \frac{\bar{\alpha}_{t-1}\beta_t^2}{(1 - \bar{\alpha}_t)^2} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[\|\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

$$= \omega_t \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[\|\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

x_0 Predictor



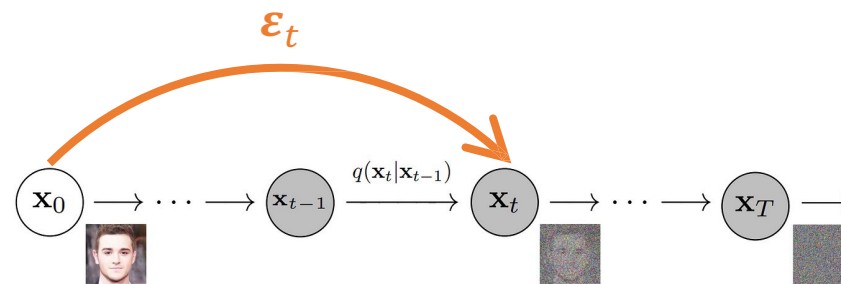
$$\omega_t \mathbb{E}_{q(x_t|x_0)} [\|\hat{x}_\theta(x_t, t) - x_0\|^2]$$

- x_t is sampled from x_0 .

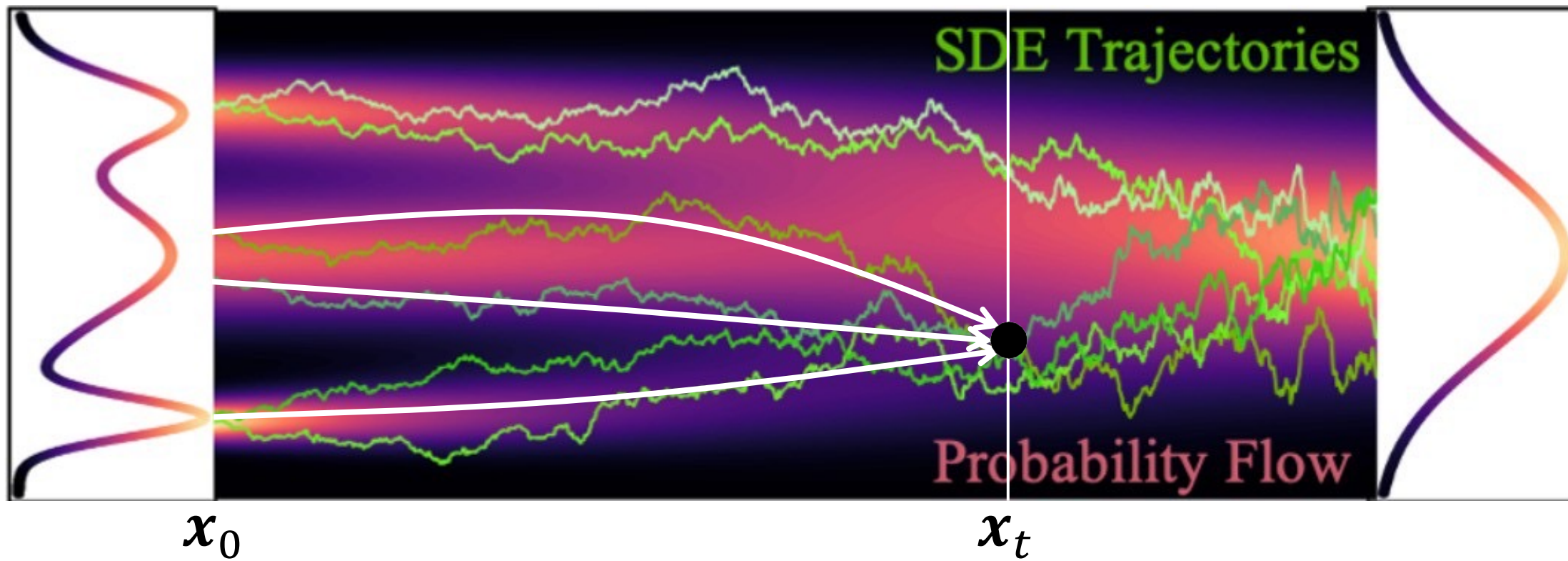
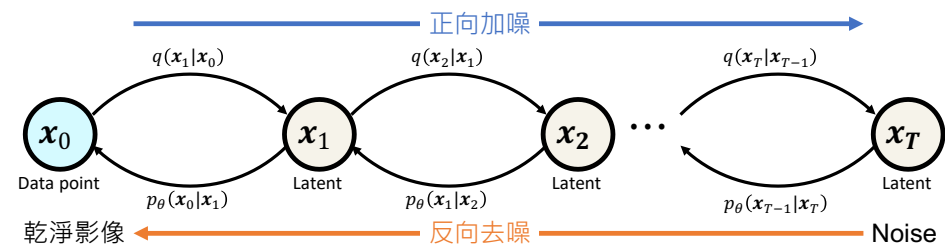
給定一張 noisy 影像 x_t ，理論上有無數個可能的原始影像 x_0 都能通過加 noise 產生這個 x_t

- From x_t , predict the *expected* value of x_0 that would result in sampling x_t from it through the forward jump.

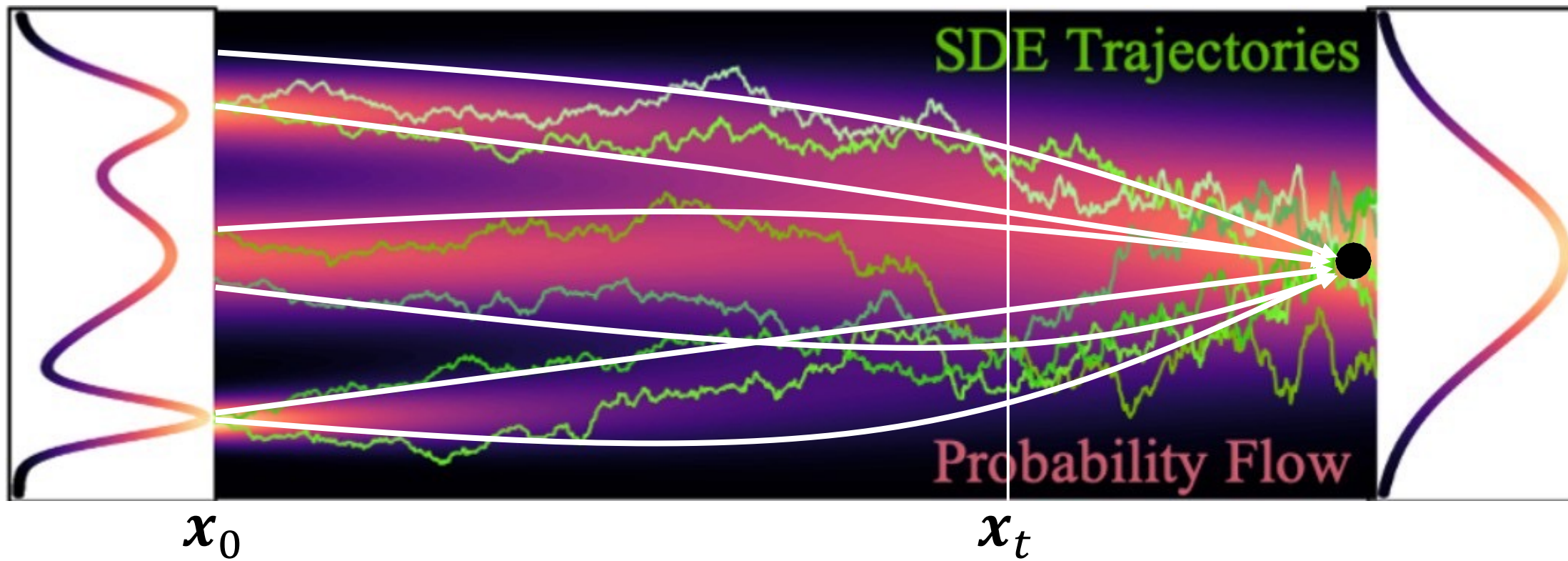
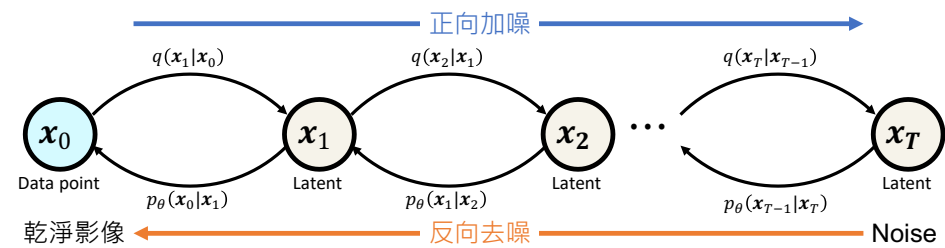
x_0 Predictor 在做的事情是：看到一張 noisy 影像 x_t ，猜測原始影像 x_0 ，但因為有多種可能，所以預測「平均來說最可能」（期望值）的那個



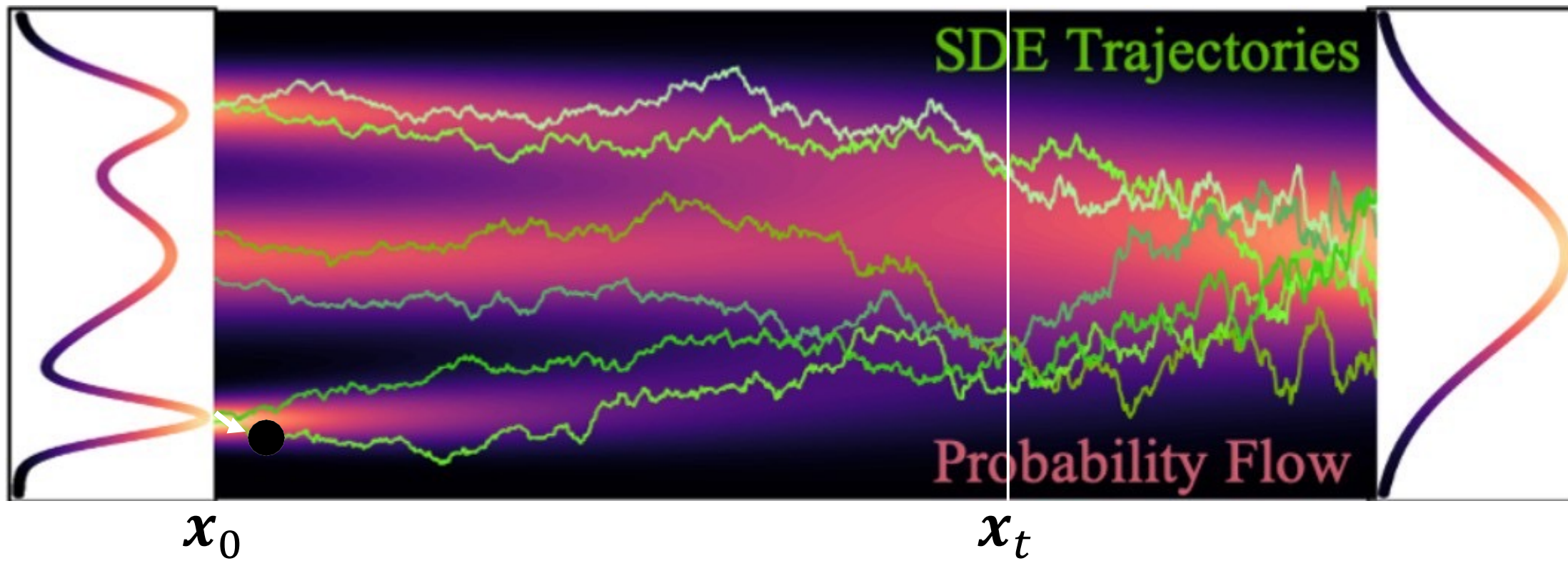
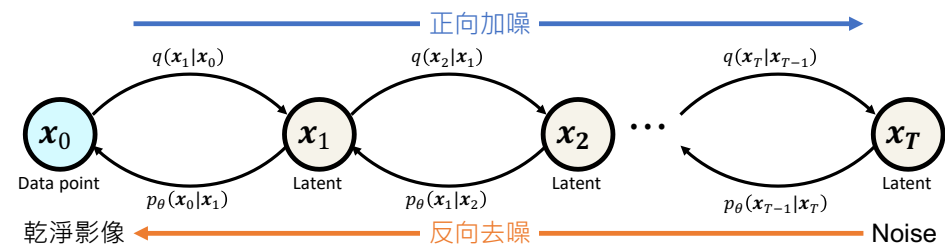
x_0 Predictor



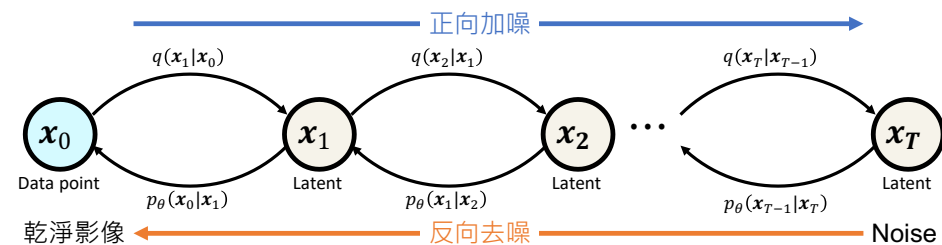
x_0 Predictor



x_0 Predictor



x_0 Predictor



- Note that our goal is to sample x_0 from a standard normal sample x_T and through latent variables $x_{T-1}, x_{T-2}, \dots, x_1$.
- But for every x_t , we directly predict the expected value of x_0 from x_t .

雖然生成過程是逐步的，但訓練時我們其實是直接預測原始數據 x_0

- 表面上看起來像是每步只預測「前一步」
- 實際上在每個 time step t ，network 都在嘗試直接預測 x_0

為什麼這樣設計？

因為 posterior 的 mean 依賴於 x_0 : $q(x_{t-1}|x_t, x_0) = \mathcal{N}(\tilde{\mu}(x_t, x_0), \tilde{\sigma}_t^2 \mathbf{I})$

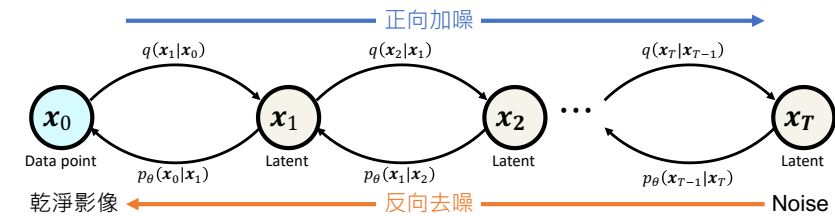
(x_{t-1} 的 mean $\tilde{\mu}$ 是 x_t 與 x_0 的線性內差) 所以需要預測 x_0

不是盲目地去噪，而是基於對最終目標的理解這種「**全局視野**」，每一步都在糾正對原始圖像的估計，讓模型能更好地保持圖像的結構

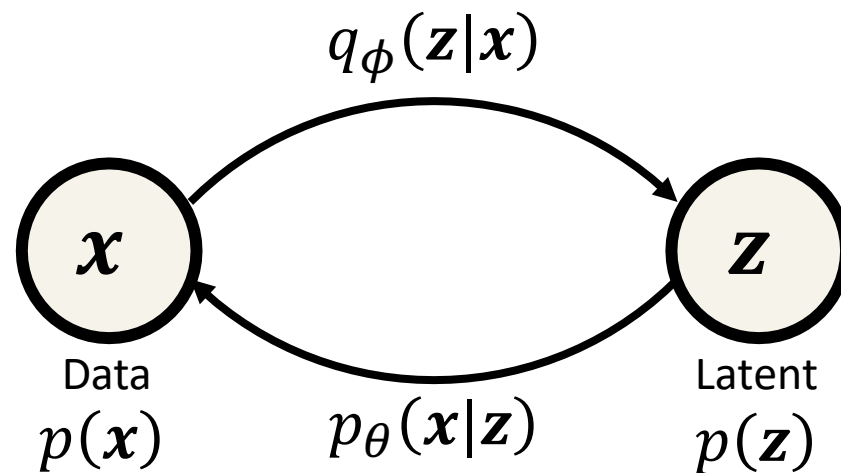
Denoising Diffusion Probabilistic Models (DDPM)

Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020.

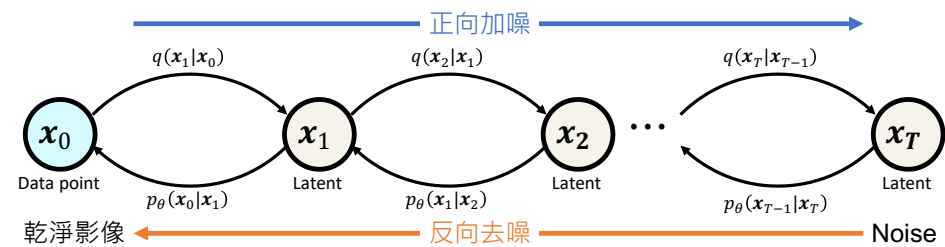
Basic Idea of Generative Models



$$\text{Posterior Encoder } p(\mathbf{z}|\mathbf{x}) = \frac{\text{Likelihood Decoder } p(\mathbf{x}|\mathbf{z}) \text{ Prior Latent } p(\mathbf{z})}{\text{Marginal Data } p(\mathbf{x})}$$

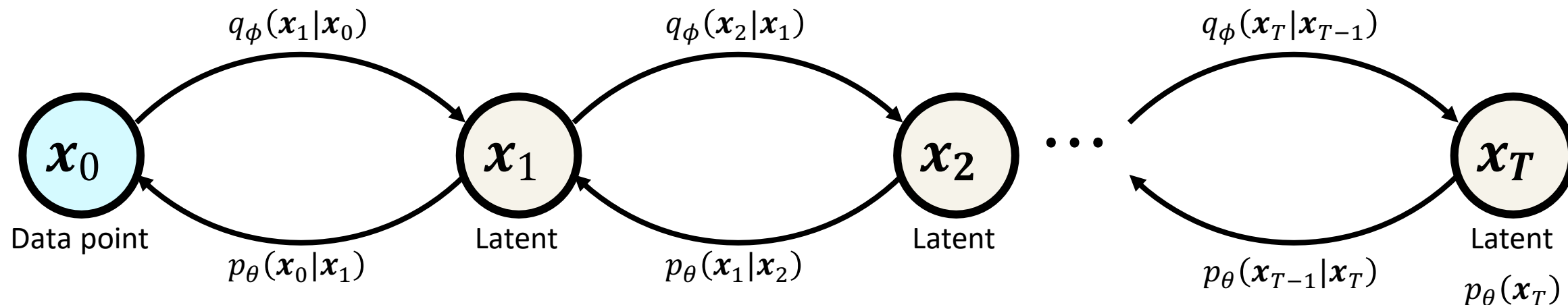


Variational Diffusion Models

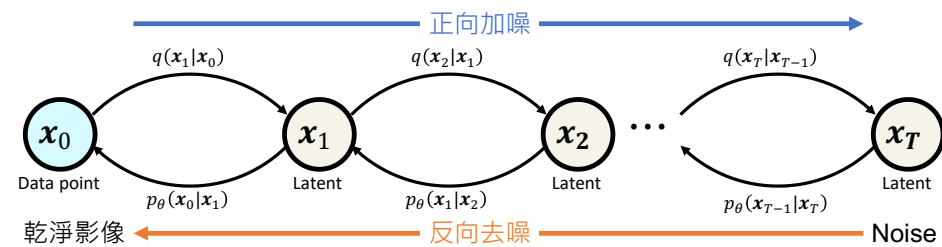


Key features:

- Both forward (encoding) and reverse (decoding) processes are **sequential** processes (with a sequence of latent variables).

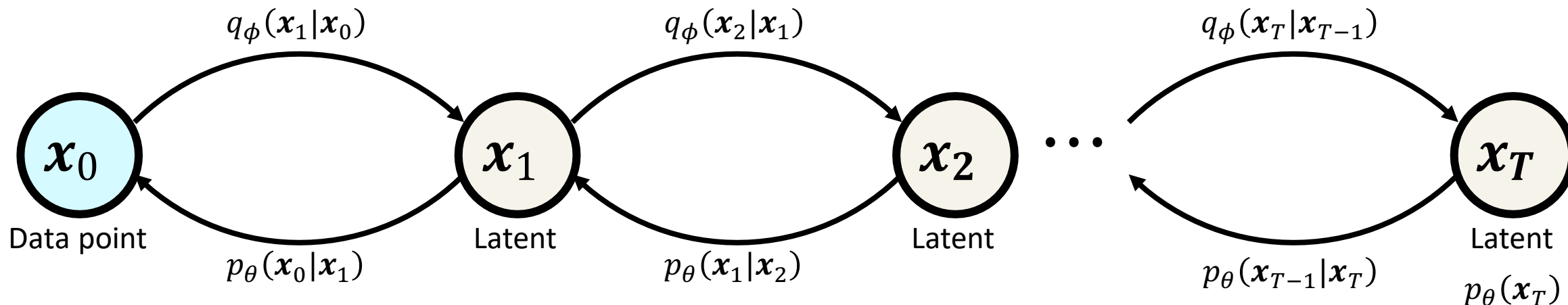


Variational Diffusion Models

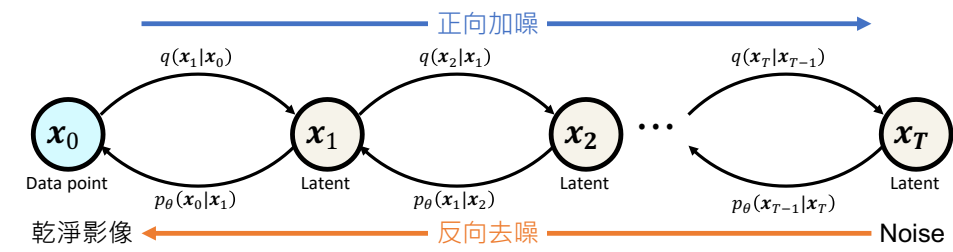


Key features:

1. Both forward (encoding) and reverse (decoding) processes are **sequential** processes (with a sequence of latent variables).
2. The **forward process** is not learned but **predefined**.



Variational Diffusion Models



Key features:

1. Both forward (encoding) and reverse (decoding) processes are **sequential** processes (with a sequence of latent variables).
2. The **forward process** is not learned but **predefined**.
3. The **dimensions** of the latent variables and the input data are the **same**.

Denoising Diffusion Probabilistic Models (DDPM)

Key features of DDPM (specifically):

1. The sequential forward and reverse process are **Markovian** processes:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0)$$

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$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0)$$

2. The **forward transitional distribution** is defined as follows:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

where $\{\beta_t \in (0,1)\}_{t=1}^T$ and $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$.

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Q. 為什麼正向加噪要這樣定義？

Denoising Diffusion Probabilistic Models (DDPM)

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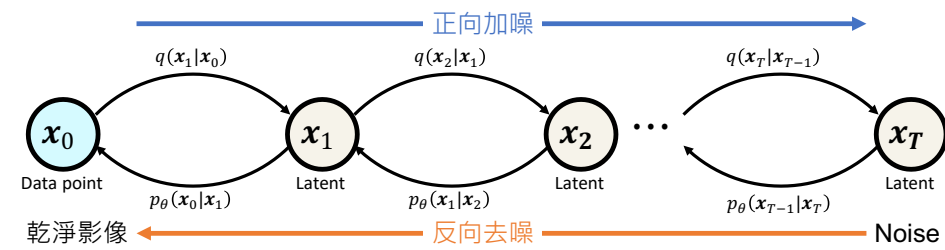
where $\{\beta_t \in (0,1)\}_{t=1}^T$ and $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$.

- A.** 保證收斂到 standard normal distribution , $q(\mathbf{x}_T | \mathbf{x}_0) \rightarrow \mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$

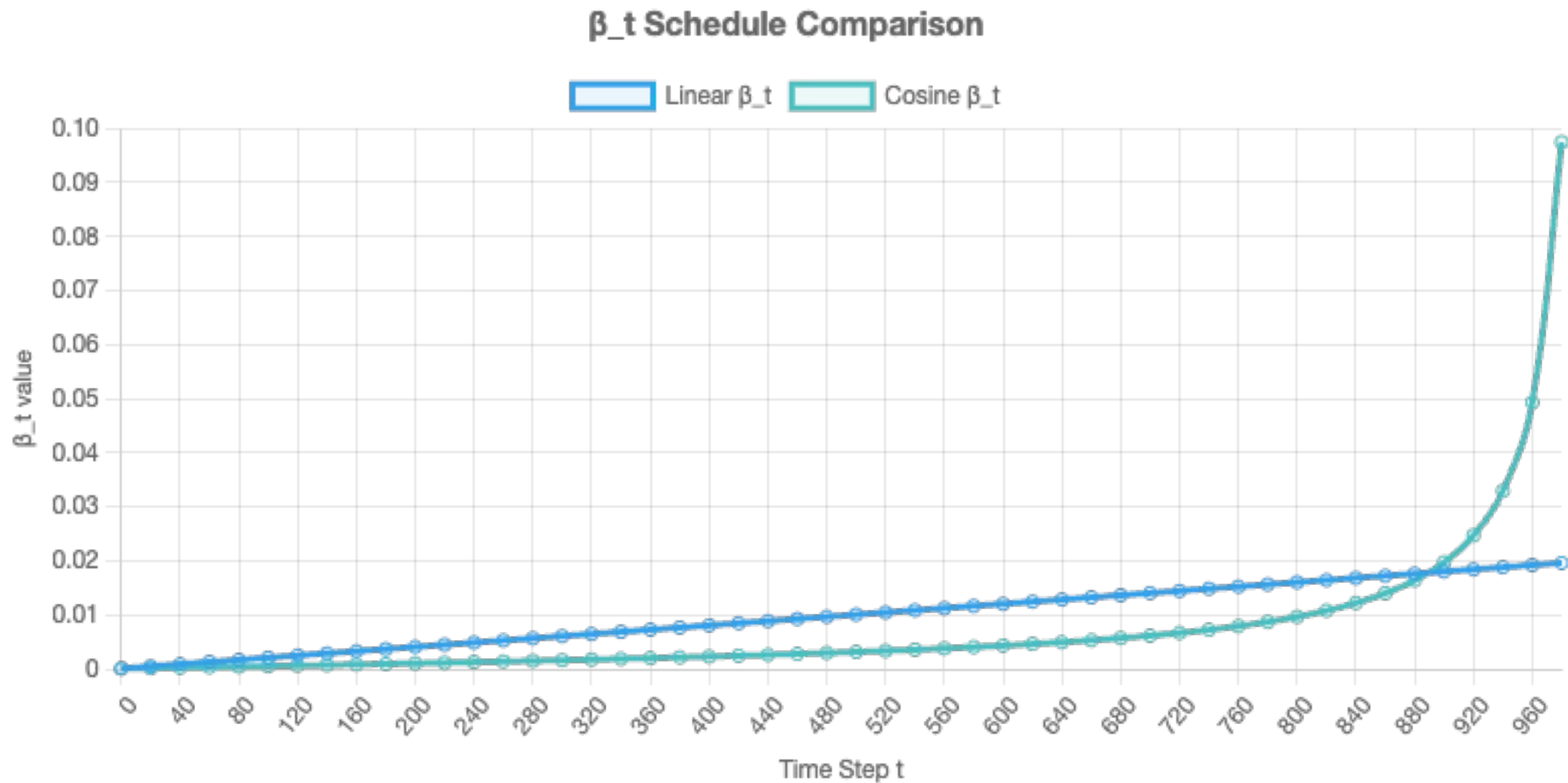
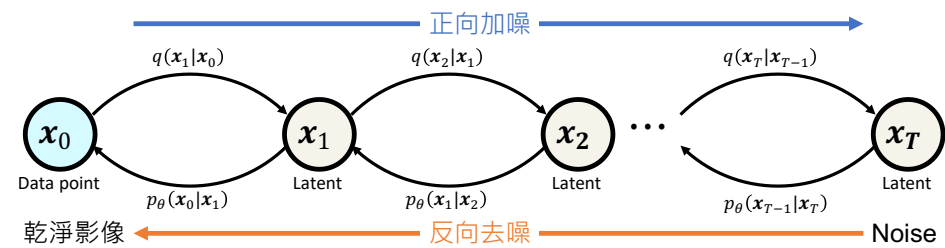
$$\mathbf{x}_T \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Choice of β_t

- Learned.
- Constant.
- Linearly or quadratically increased.
- Follows a **cosine** function
(Nichol and Dhariwal, Improved Denoising Diffusion Probabilistic Models, ICML 2021).
- Note that the reverse step $p_\theta(x_{t-1}|x_t)$ becomes a Gaussian form only when β_t is small ($\beta_t \ll 1$).



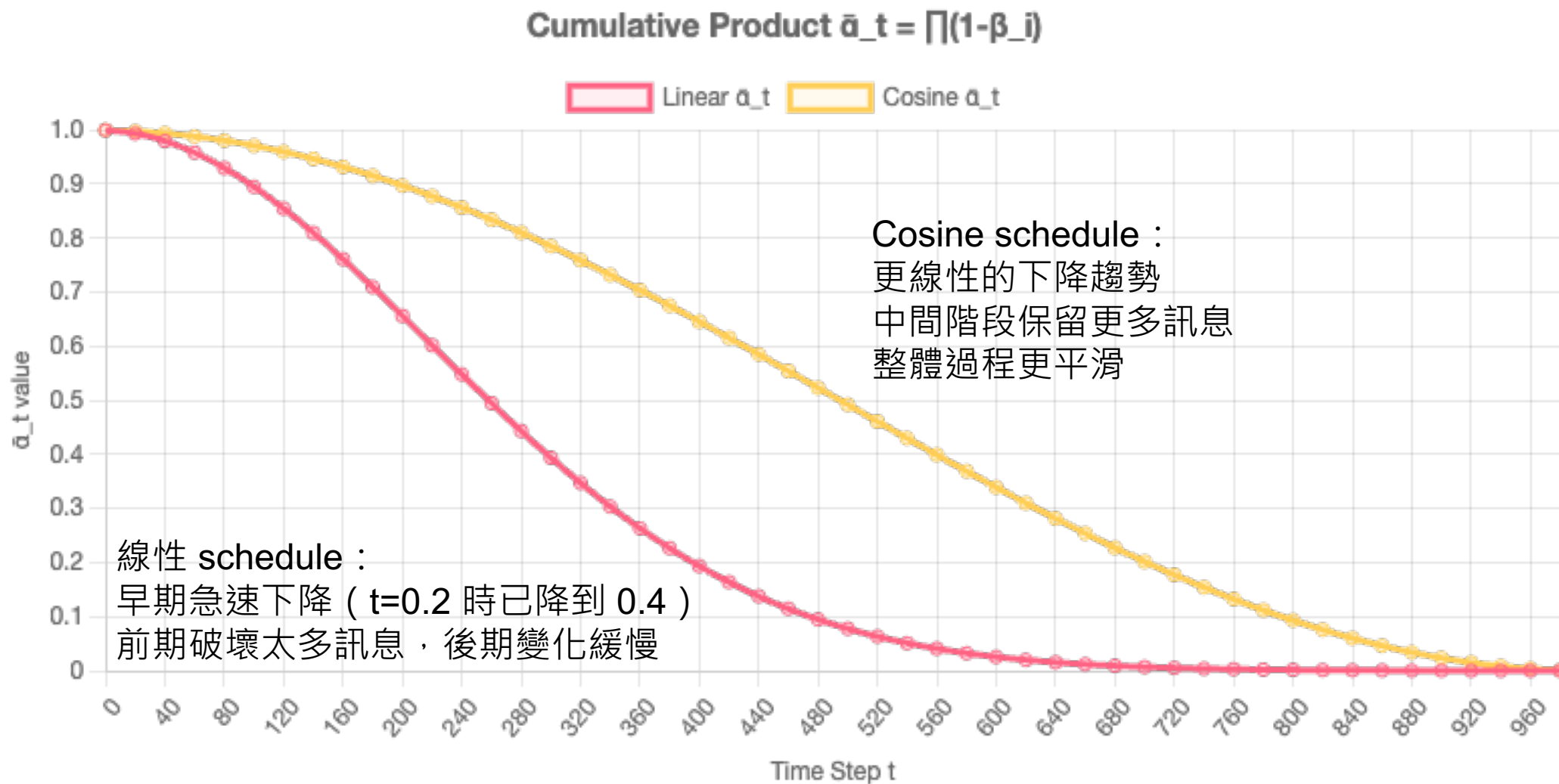
Choice of β_t



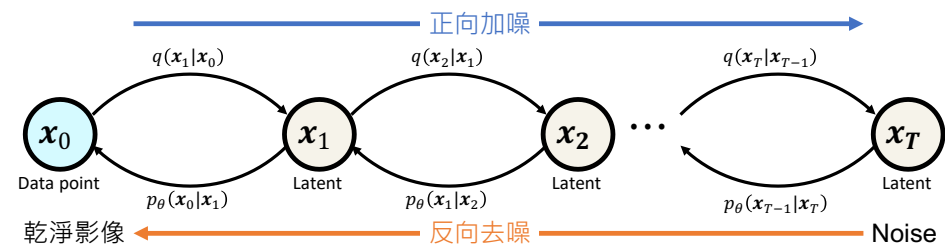
Choice of β_t

$\bar{\alpha}_t$ 代表原始影像的保留比例

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$



Choice of β_t



Linear Schedule

$$\beta_t = 0.0001 + \frac{t}{T} (0.02 - 0.0001)$$

簡單直接的線性增長，早期階段 noise 增加較快，可能導致訓練不穩定

Cosine Schedule

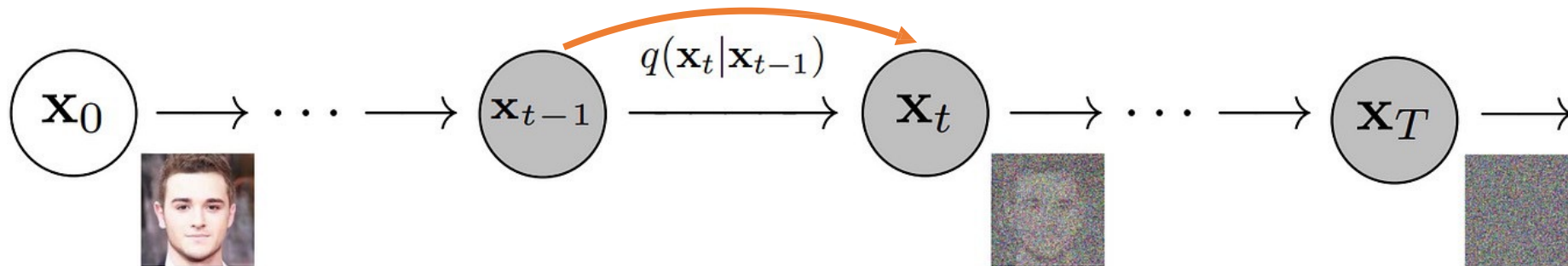
$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, f(t) = \cos \left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2} \right)^2$$

更平滑的噪音增加，中間階段保留更多資訊，訓練更穩定，生成品質通常更好，適合高解析度影像

Denoising Diffusion Probabilistic Models (DDPM)

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

where $\alpha_t = 1 - \beta_t$.

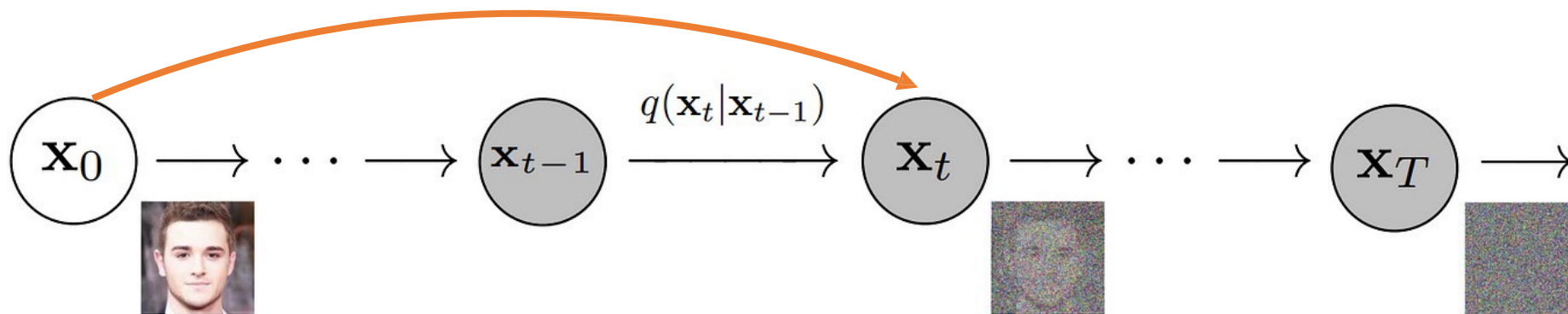


Denoising Diffusion Probabilistic Models (DDPM)

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

串起來推導可得

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

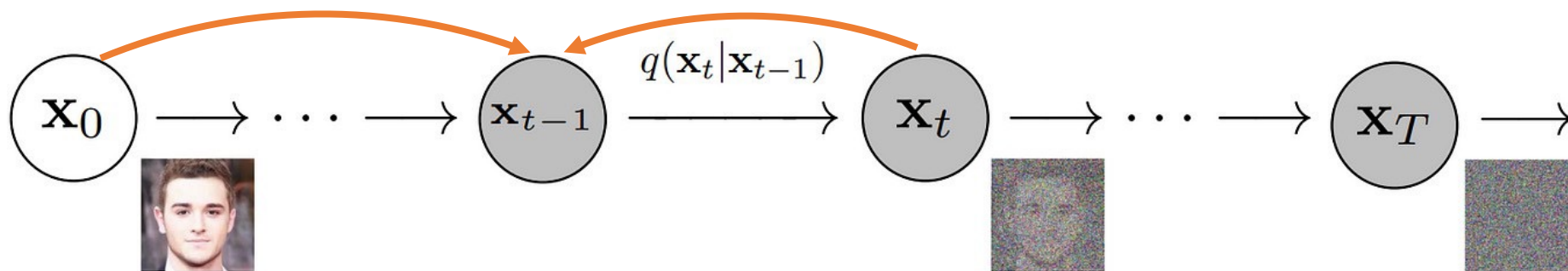


Denoising Diffusion Probabilistic Models (DDPM)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

where $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0$ and $\tilde{\sigma}_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t$.

$\mathbf{x}_{t-1}$ 的 mean $\tilde{\mu}$ 是 $\mathbf{x}_t$ 與 $\mathbf{x}_0$ 的線性內差



Denoising Diffusion Probabilistic Models (DDPM)

1. $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$
where $\alpha_t = 1 - \beta_t$.

正向加噪過程，是我們精心定義的

2. $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$
where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

正向跳躍，串起來推導可得

3. $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$
where $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{x}_0$ and $\tilde{\sigma}_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$.

$\mathbf{x}_{t-1}$ 是 Gaussian, mean 是 $\mathbf{x}_t$ 與 $\mathbf{x}_0$ 的線性內差，variance 由 time step t 與 β_t 預先定義

Denoising Diffusion Probabilistic Models (DDPM)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

$$1. \quad \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0 \text{ or}$$

mean 可以由 $\mathbf{x}_t$ 與 $\mathbf{x}_0$ 線性內差得到

$$2. \quad \tilde{\mu}(\mathbf{x}_t, \boldsymbol{\varepsilon}_t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\varepsilon}_t \right)$$

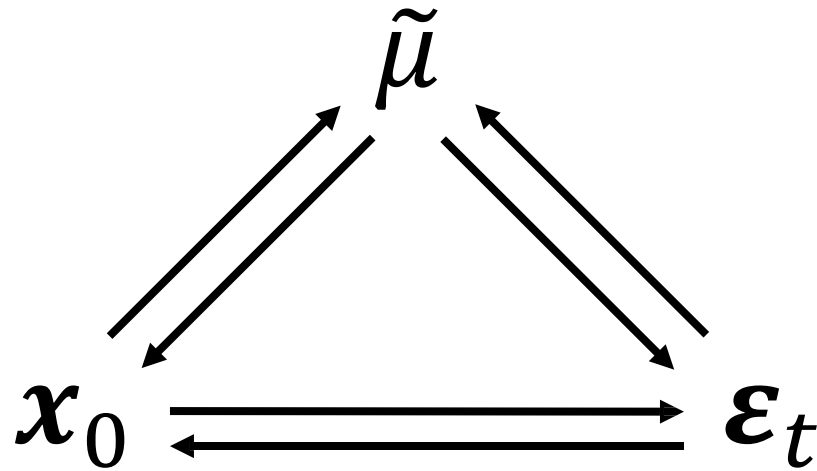
$$\text{where } \boldsymbol{\varepsilon}_t = \frac{1}{\sqrt{1-\bar{\alpha}_t}} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_t}}{\sqrt{1-\bar{\alpha}_t}} \mathbf{x}_0$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t} \boldsymbol{\varepsilon}_t$$

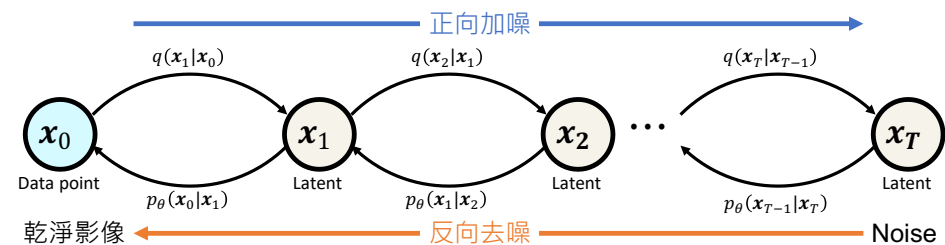
mean 也可以由 $\mathbf{x}_t$ 與 noise $\boldsymbol{\varepsilon}_t$ 得到

Denoising Diffusion Probabilistic Models (DDPM)

Given x_t , each of $\tilde{\mu}$, x_0 , and ϵ_t can be computed from any of the others.



ELBO



The negative ELBO is decomposed into three terms:

$$-\mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] = \dots =$$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

Reconstruction
term $\mathcal{L}_0$

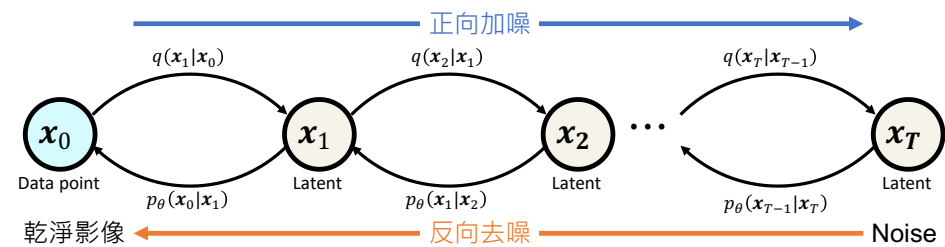
$$+D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T))$$

Prior matching
term $\mathcal{L}_T$

$$+\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Denoising
matching
term $\mathcal{L}_{t-1}$

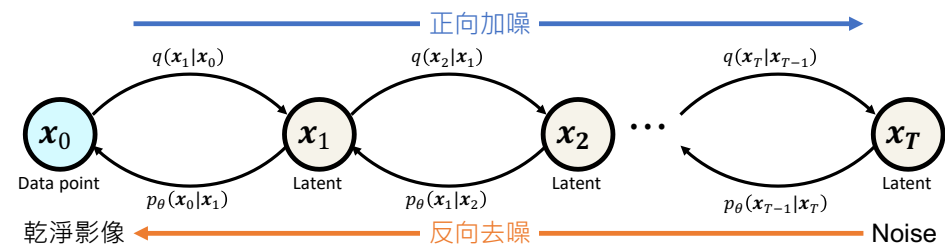
ELBO



- Reconstruction term $\mathcal{L}_0 = -\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$: The same as with VAEs; **it is also negligible**.

Q. 為什麼這一項可以忽略？

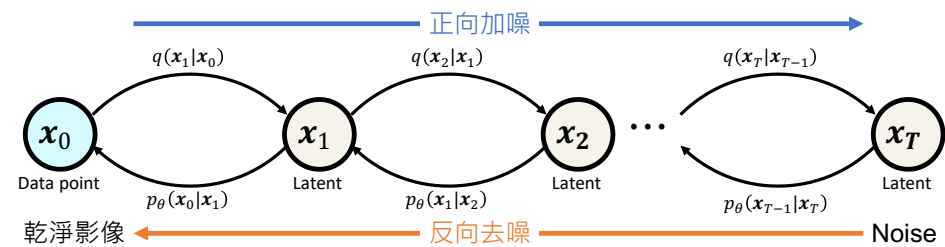
ELBO



- Reconstruction term $\mathcal{L}_0 = -\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$: The same as with VAEs; **it is also negligible**.

A. 因為 β_1 通常很小，所以 $\mathbf{x}_1 \approx \mathbf{x}_0$ ，重建變得微不足道，且相對於其他項很小

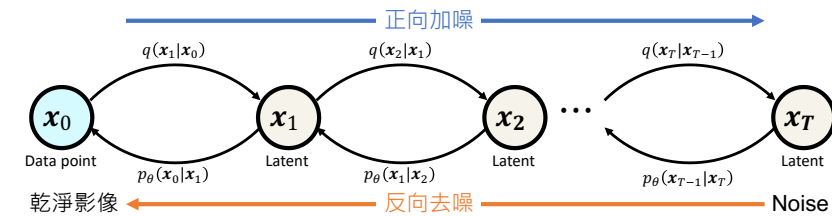
ELBO



- Reconstruction term $\mathcal{L}_0 = -\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$: The same as with VAEs; **it is also negligible**.
- Prior matching term $\mathcal{L}_T = D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))$: **Converges to zero** when $T \rightarrow \infty$.
- Denoising matching term $\mathcal{L}_{t-1}$

$$\mathcal{L}_{t-1} = \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Denoising Matching Term $\mathcal{L}_{t-1}$



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))] \\ = \mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

How to model the variational distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$?

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

Three options to define variational distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

1. $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I})$

with the mean predictor $\mu_{\theta}(\mathbf{x}_t, t)$.

希望 variational distribution 的 mean 與 posterior distribution 的 mean 越像越好

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

Three options to define variational distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

$$2. \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

with the $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$.

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

Three options to define variational distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

$$3. \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\boldsymbol{\varepsilon}}_t(\mathbf{x}_t, t)\right), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

with the $\boldsymbol{\varepsilon}_t$ predictor $\hat{\boldsymbol{\varepsilon}}_{\theta}(\mathbf{x}_t, t)$.

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

1. Mean predictor $\mu_{\theta}(\mathbf{x}_t, t)$:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I})$$

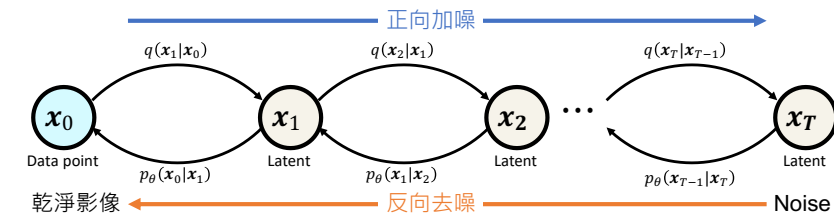
2. $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

3. $\boldsymbol{\varepsilon}_t$ predictor $\hat{\boldsymbol{\varepsilon}}_{\theta}(\mathbf{x}_t, t)$:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\boldsymbol{\varepsilon}}_{\theta}(\mathbf{x}_t, t) \right), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

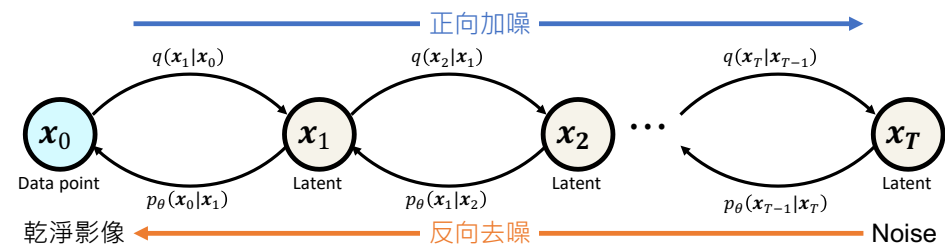
Denoising Matching Term $\mathcal{L}_{t-1}$



$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

How can we minimize the denoising matching term for each predictor?

Diffusion Matching Term



Rewrite the diffusion matching term for each case:

1. Mean predictor $\mu_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\sigma}_t^2} \|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2 \right]$$

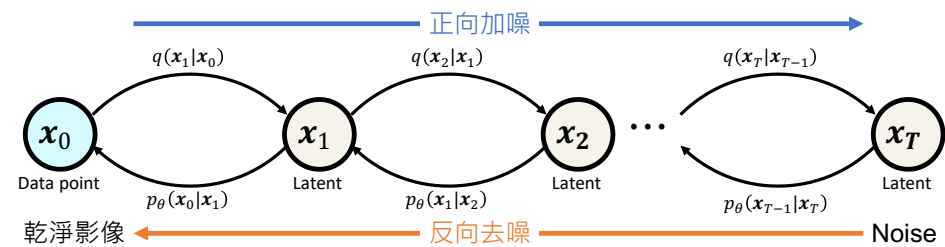
2. $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\sigma}_t^2} \omega_t \|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2 \right]$$

3. $\boldsymbol{\varepsilon}_t$ predictor $\hat{\boldsymbol{\varepsilon}}_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\sigma}_t^2} \omega'_t \|\hat{\boldsymbol{\varepsilon}}_\theta(\mathbf{x}_t, t) - \boldsymbol{\varepsilon}_t\|^2 \right]$$

Diffusion Matching Term



In practice, we can simply **drop the weight term** in training:

1. Mean predictor $\mu_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [\|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2]$$

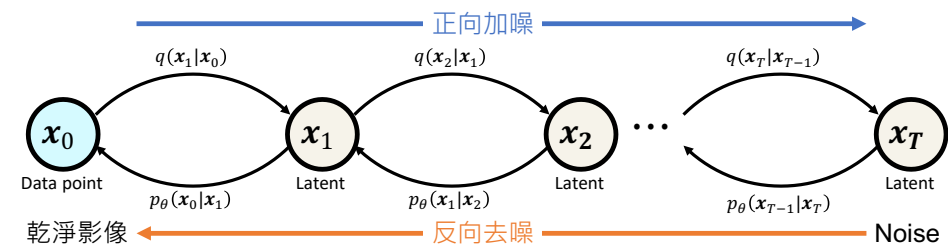
2. $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

3. $\boldsymbol{\varepsilon}_t$ predictor $\hat{\boldsymbol{\varepsilon}}_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [\|\hat{\boldsymbol{\varepsilon}}_\theta(\mathbf{x}_t, t) - \boldsymbol{\varepsilon}_t\|^2]$$

Diffusion Matching Term



In practice, we generally use the ϵ_t predictor $\hat{\epsilon}_\theta(x_t, t)$ since

ϵ_t are standard normal samples

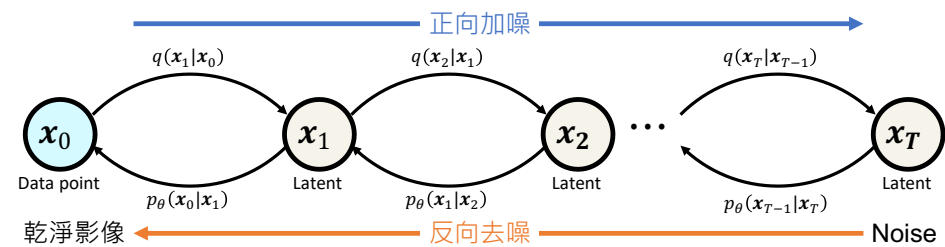
➔ well normalized and scaled

➔ which makes it easier to train a neural network.

數值穩定：噪音的範圍固定在 $\mathcal{N}(\mathbf{0}, \mathbf{I})$

效果最好：實驗證明這種參數化表現最佳

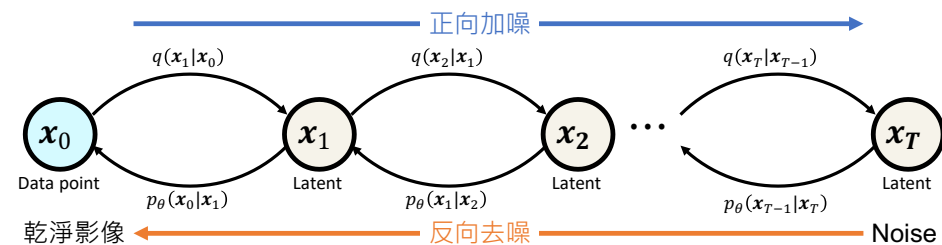
Training



How to train the ϵ_t predictor $\hat{\epsilon}_\theta(\mathbf{x}_t, t)$ with the loss function:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [\|\hat{\epsilon}_\theta(\mathbf{x}_t, t) - \epsilon_t\|^2]$$

Training



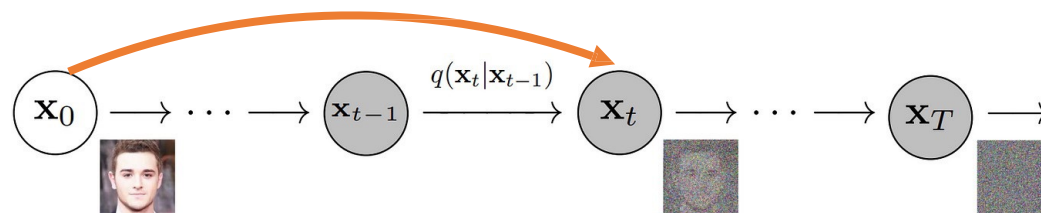
$$\mathbb{E}_{t>1, q(x_t|x_0)} [\|\hat{\epsilon}_\theta(x_t, t) - \epsilon_t\|^2]$$

Repeat:

1. Take a random x_0 .
2. Sample $t \sim \mathcal{U}(\{1, \dots, T\})$.
3. Sample $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t$.
5. Take gradient decent step on $\nabla_\theta \|\hat{\epsilon}_\theta(x_t, t) - \epsilon_t\|^2$.

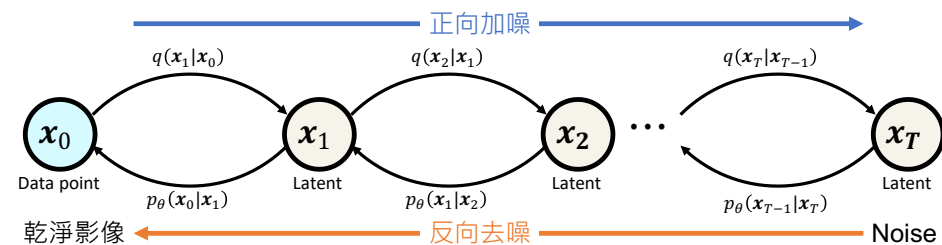
Same as sampling

$$x_t \sim q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, \sqrt{1 - \bar{\alpha}_t} \mathbf{I})$$



Training data: 用噪音去淹沒圖片內容
Network 需要學會正確預測噪音

Training



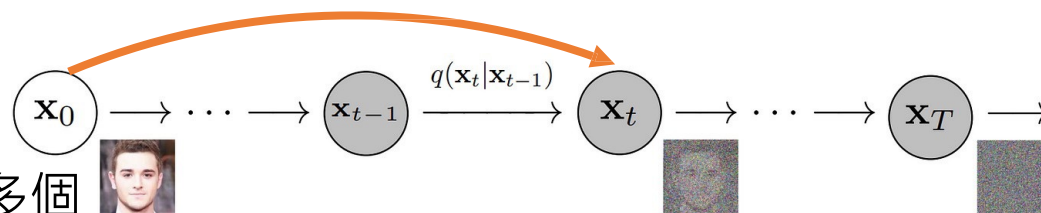
$$\mathbb{E}_{t>1, q(x_t|x_0)} [\|\hat{\epsilon}_\theta(x_t, t) - \epsilon_t\|^2]$$

Repeat:

1. Take a random x_0 .
2. Sample $t \sim \mathcal{U}(\{1, \dots, T\})$.
3. Sample $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t$.
5. Take gradient decent step on $\nabla_\theta \|\hat{\epsilon}_\theta(x_t, t) - \epsilon_t\|^2$.

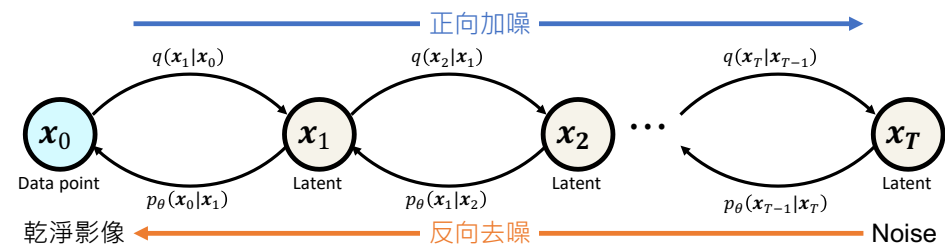
Same as sampling

$$x_t \sim q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, \sqrt{1 - \bar{\alpha}_t} \mathbf{I})$$



並不是讓 network 去死背 noise，因為可能有很多個 x_0 在正向加噪跳躍會變成一樣的 x_t ，網絡學習的是條件期望的統計規律（最可能的去噪方向），而不是簡單的記憶一對一映射

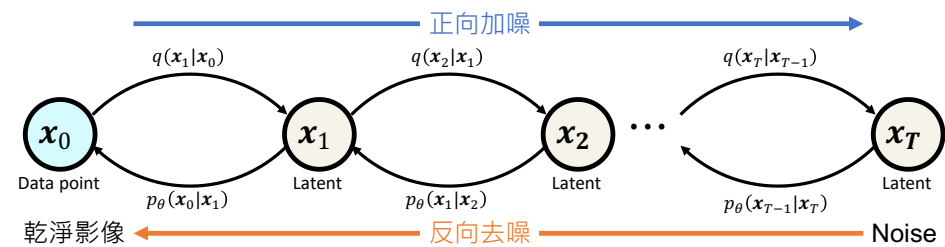
Reverse Process (Generation)



How to proceed the reverse process with the **learned** reverse transitional distribution:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t)\right), \tilde{\sigma}_t^2\mathbf{I}\right)$$

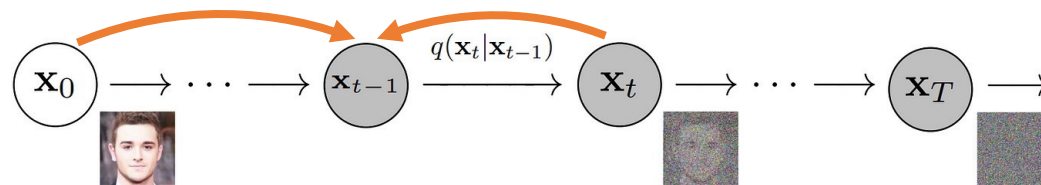
Reverse Process (Generation)



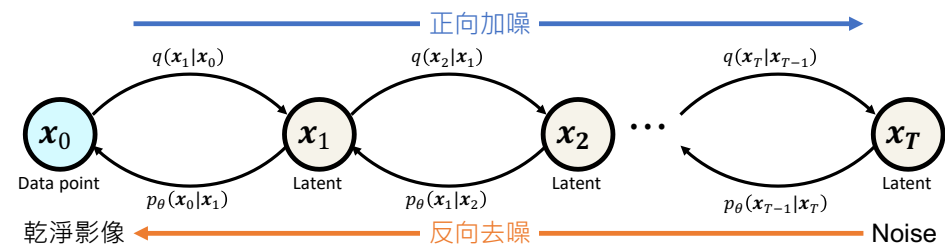
1. Sample $\mathbf{x}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. For $t = T, \dots, 1$, repeat:

1. Compute $\tilde{\mu} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \hat{\mathbf{e}}_\theta(\mathbf{x}_t, t) \right)$.
2. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
3. Compute $\mathbf{x}_{t-1} = \tilde{\mu} + \tilde{\sigma} \mathbf{z}_t$.

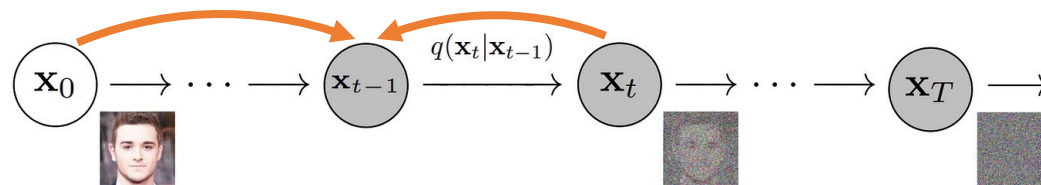
Same as sampling
 $\mathbf{x}_t \sim p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$.



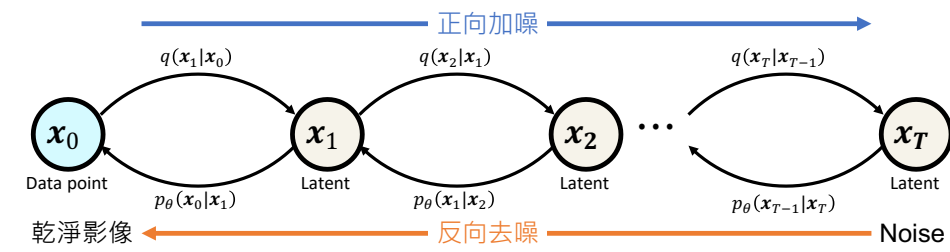
Reverse Process (Generation)



1. Sample $\mathbf{x}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. For $t = T, \dots, 1$, repeat:
 1. Compute $\tilde{\mu} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t) \right)$.
 2. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
 3. Compute $\mathbf{x}_{t-1} = \tilde{\mu} + \tilde{\sigma} \mathbf{z}_t$.



Reverse Process (Generation)

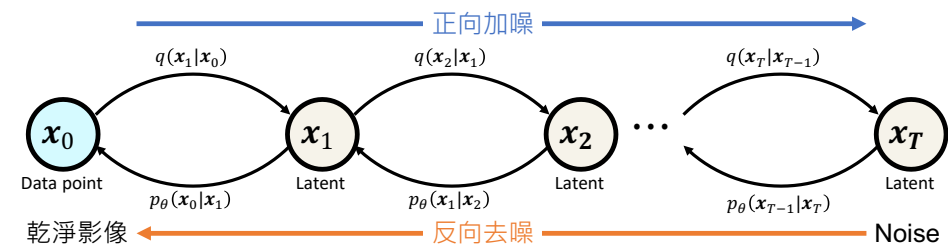


At each timestep t , given x_t ,

- ϵ_t is predicted.
- The prediction of x_0 can be computed from x_t and ϵ_t .

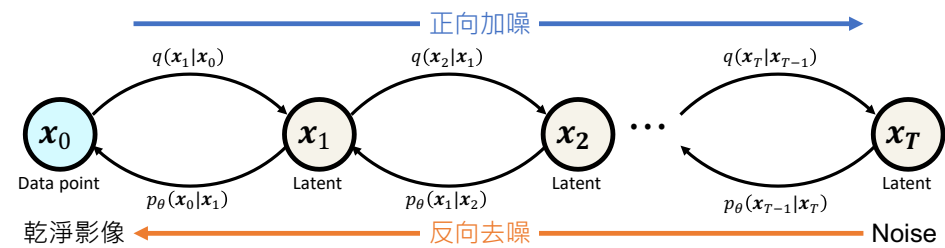
The denoising (reverse) process can also be viewed as a refinement process!

Reverse Process (Generation)



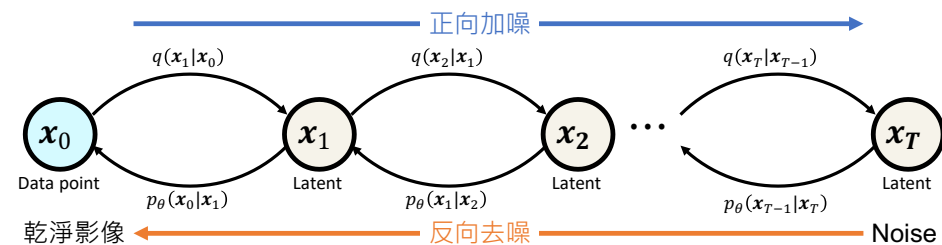
Q. Given x_t and the noise prediction $\hat{\epsilon}_\theta(x_t, t)$, what is the prediction of x_0 (which will be denoted as $x_{0|t}$)?

Reverse Process (Generation)



$$\mathbf{A. } \mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t) \right)$$

Reverse Process (Generation)



Reverse Process with ε_t Predictor

1. Sample $\mathbf{x}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. For $t = T, \dots, 1$, repeat:

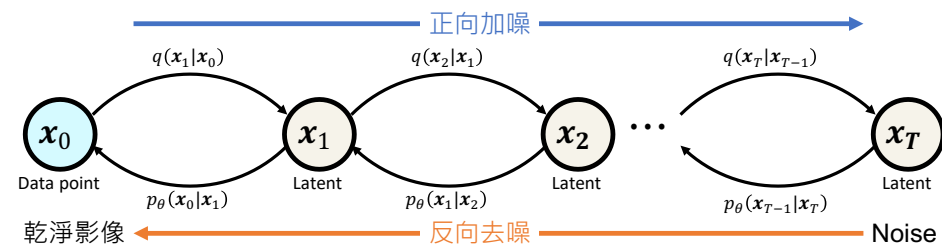
1. Compute $\mathbf{x}_{0|t}$ from $\mathbf{x}_t$ and $\hat{\varepsilon}_\theta(\mathbf{x}_t, t)$.
2. Compute $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_{0|t})$.
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\mu} + \tilde{\sigma} \mathbf{z}_t$.

跟前面的 reverse process 不太一樣
得到 noise prediction $\hat{\varepsilon}_\theta$ 之後，不是直接扣掉
而是 (1) 先用 $\mathbf{x}_t$ 跟 $\hat{\varepsilon}_\theta$ 外推出 $\mathbf{x}_{0|t}$
再 (2) 計算出 mean of the likelihood distribution $\tilde{\mu}$

Same as sampling
 $\mathbf{x}_t \sim p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$.

這就是他的 mean!
不要被下標 $t, t-1$ 混淆了

Recap of All the Steps



ELBO

Decompose the negative ELBO in a **different** way:

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$$= \dots =$$

$-\mathbb{E}_{q(\mathbf{x}_1 \mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0 \mathbf{x}_1)]$	Reconstruction term $\mathcal{L}_0$
$+D_{KL}(q(\mathbf{x}_T \mathbf{x}_0) \ p(\mathbf{x}_T))$	New prior matching term $\mathcal{L}_T$
$+ \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t \mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1} \mathbf{x}_t, \mathbf{x}_0) \ p_\theta(\mathbf{x}_{t-1} \mathbf{x}_t))]$	Denoising matching term $\mathcal{L}_{t-1}$

Denoising Matching Term $\mathcal{L}_{t-1}$

$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

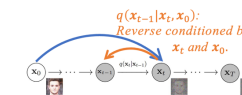
The **variational** distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ should be close to $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ for each t .

$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

- The **mean** $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0$ is a function of both $\mathbf{x}_t$ and $\mathbf{x}_0$.
 $\mathbf{x}_{t-1}$ 的 mean $\tilde{\mu}$ 是 $\mathbf{x}_t$ 與 $\mathbf{x}_0$ 的線性內差

- The **covariance** $\tilde{\sigma}_t^2 \mathbf{I} = \left(\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t \right) \mathbf{I}$ is **predefined** from the user-defined $\{\beta_t\}_{t=1}^T$.

$\mathbf{x}_{t-1}$ 的 variance $\tilde{\sigma}_t^2$ 與 $\mathbf{x}_t$ 或 $\mathbf{x}_0$ 無關。僅由 time step t 與 β_t 預先定義



Denoising Diffusion Probabilistic Models (DDPM)

1. $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$
where $\alpha_t = 1 - \beta_t$. 正向加噪過程，是我們精心定義的

2. $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$
where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$. 正向跳躍，串起來推導可得

3. $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$
where $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_t}(1-\bar{\alpha}_{t-1})}{\sqrt{1-\bar{\alpha}_t}} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{\sqrt{1-\bar{\alpha}_t}} \mathbf{x}_0$ and $\tilde{\sigma}_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t$.

$\mathbf{x}_{t-1}$ 是 Gaussian, mean 是 $\mathbf{x}_t$ 與 $\mathbf{x}_0$ 的線性內差，variance 由 time step t 與 β_t 預先定義

Denoising Diffusion Probabilistic Models (DDPM)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

1. $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0$ or
mean 可以由 $\mathbf{x}_t$ 與 $\mathbf{x}_0$ 線性內差得到

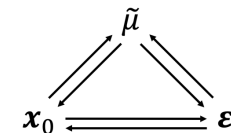
2. $\tilde{\mu}(\mathbf{x}_t, \boldsymbol{\varepsilon}_t) = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\varepsilon}_t \right)$
where $\boldsymbol{\varepsilon}_t = \frac{1}{\sqrt{1-\bar{\alpha}_t}} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_t}}{\sqrt{1-\bar{\alpha}_t}} \mathbf{x}_0$

mean 也可以由 $\mathbf{x}_t$ 與 noise $\boldsymbol{\varepsilon}_t$ 得到

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t} \boldsymbol{\varepsilon}_t$$

Denoising Diffusion Probabilistic Models (DDPM)

Given $\mathbf{x}_t$, each of $\tilde{\mu}$, $\mathbf{x}_0$, and $\boldsymbol{\varepsilon}_t$ can be computed from any of the others.



Diffusion Matching Term

$$\begin{aligned} & \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_t \sim p_\theta} \left[\mathbb{E}_{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \left[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) \right] \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_t \sim p_\theta} \left[\mathbb{E}_{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \left[\frac{1}{2} \left\| \frac{\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\tilde{\sigma}_t} \right\|^2 \right] \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_t \sim p_\theta} \left[\frac{1}{2} \left\| \frac{\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\tilde{\sigma}_t} \right\|^2 \right] \end{aligned}$$

Direct Sampling vs. Generative Models

Q. Why don't we directly sample from the training data instead of training a generative model?



Direct Sampling vs. Generative Models

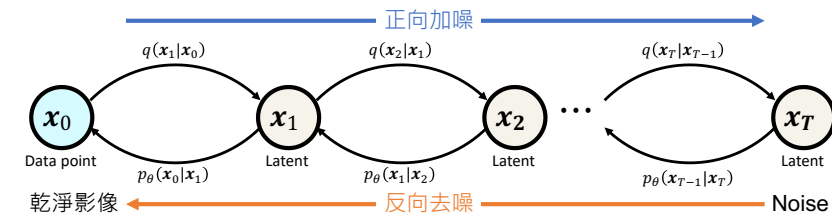
- **Efficiency and Scalability:**
Generative models can efficiently encode the data distribution into a neural network, eliminating the need to store the entire dataset.
- **Novelty and Diversity / Privacy and Security Concerns:**
Generative models can generate new samples that are not identical to any specific training data but still appear plausible.
- **Adaptability to New Conditions:**
For conditional generation, generative models can be more efficient at generating data based on the given conditions compared to retrieval methods.

Reverse Process (Generation)



在每個 time step t ，network 都在嘗試直接預測 x_0
不是盲目地去噪，而是基於對最終目標的理解這種「**全局視野**」，每一步都在糾正對原始圖像的估計，讓模型能更好地保持圖像的結構

Denoising Matching Term $\mathcal{L}_{t-1}$



$$\sum_{t=2}^T \mathbb{E}_{q(x_t|x_0)} [D_{KL}(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t))] \\ = \mathbb{E}_{t>1, q(x_t|x_0)} [D_{KL}(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t))]$$

How to model the variational distribution $p_\theta(x_{t-1}|x_t)$?

1st Assignment

